

IEEE Standard for Local and metropolitan area networks— Bridges and Bridged Networks— Amendment 28: Per-Stream Filtering and Policing

IEEE Computer Society

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(Amendment to
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as amended by
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IEEE Std 802.1Qcd™-2015,
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IEEE Std 802.1Qbz™-2016)

**IEEE Standard for
Local and metropolitan area networks—**

Bridges and Bridged Networks—

Amendment 28: Per-Stream Filtering and Policing

Sponsor

**LAN/MAN Standards Committee
of the
IEEE Computer Society**

Approved 14 February 2017

IEEE-SA Standards Board

Abstract: Enhancements to the forwarding process that support per-stream filtering and policing are provided in this amendment to IEEE Std 802.1Q-2014.

Keywords: Bridged Local Area Networks, IEEE 802[®], IEEE 802.1Q[™], IEEE Std 802.1Qbu[™], IEEE 802.1Qbv[™], IEEE Std 802.1Qbz[™], IEEE 802.1Qca[™], IEEE 802.1Qcd[™], IEEE 802.1Qci[™], local area networks (LANs), MAC Bridges, metropolitan area networks, per-stream filtering and policing, PSFP, scheduled traffic, Time-Sensitive Networking, Virtual Bridged Local Area Networks (virtual LANs)

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Introduction

This introduction is not part of IEEE Std 802.1Qci-2017, IEEE Standard for Local and metropolitan area networks—Bridges and Bridged Networks—Amendment 28: Per-Stream Filtering and Policing.

This amendment to IEEE Std 802.1Q-2014 provides enhancements to the forwarding process that support per-stream filtering and policing.

This standard contains state-of-the-art material. The area covered by this standard is undergoing evolution. Revisions are anticipated within the next few years to clarify existing material, to correct possible errors, and to incorporate new related material. Information on the current revision state of this and other IEEE 802[®] standards may be obtained from

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IEEE Standard for Local and metropolitan area networks—

Bridges and Bridged Networks—

Amendment 28: Per-Stream Filtering and Policing

(This amendment is based on IEEE Std 802.1Q™-2014 as amended by IEEE Std 802.1Qca™-2015, IEEE Std 802.1Qcd™-2015, IEEE Std 802.1Q-2014/Cor 1-2015, IEEE Std 802.1Qbv™-2015, IEEE Std 802.1Qbu™-2016, and IEEE Std 802.1Qbz™-2016.)

NOTE—The editing instructions contained in this amendment define how to merge the material contained therein into the existing base standard and its amendments to form the comprehensive standard.

The editing instructions are shown in ***bold italic***. Four editing instructions are used: change, delete, insert, and replace. ***Change*** is used to make corrections in existing text or tables. The editing instruction specifies the location of the change and describes what is being changed by using ~~strike through~~ (to remove old material) and underscore (to add new material). ***Delete*** removes existing material. ***Insert*** adds new material without disturbing the existing material. Deletions and insertions may require renumbering. If so, renumbering instructions are given in the editing instruction. ***Replace*** is used to make changes in figures or equations by removing the existing figure or equation and replacing it with a new one. Editing instructions, change markings, and this NOTE will not be carried over into future editions because the changes will be incorporated into the base standard.¹

¹Notes in text, tables, and figures are given for information only, and do not contain requirements needed to implement the standard.

1. Overview

1.3 Introduction

Insert the following item at the end of lettered list, renumbering as necessary:

- ch) This standard specifies protocols, procedures, and managed objects that allow for the filtering and policing of individual traffic streams.

2. Normative references

Insert the following references in alphanumeric order:

IEEE Std 802.1CB™, IEEE Standard for Local and Metropolitan Area Networks—Frame Replication and Elimination for Reliability.

MEF Technical Specification 10.3 (MEF 10.3), Ethernet Service Attributes Phase 3.²

Delete the reference to MEF 10.3 from the Bibliography [B45] and amend references to it in the text accordingly.

²MEF technical specifications are available from the MEF website (<http://www.mef.net>).

4. Abbreviations

Insert the following abbreviations in alphanumeric order:

IPV internal priority value specification

PSFP per-stream filtering and policing

5. Conformance

5.4 VLAN Bridge component requirements

5.4.1 VLAN Bridge component options

Insert new subclause 5.4.1.8 as shown, renumbering as necessary:

5.4.1.8 Per-stream filtering and policing (PSFP) requirements

A VLAN Bridge component implementation that conforms to the provisions of this standard for PSFP shall:

- a) Support PSFP as specified in 8.6.5.1 and 8.6.6.1.
- b) Support the state machines for stream gate control as specified in 8.6.10.
- c) Support the management entities for PSFP as specified in 12.31.

5.13 MAC Bridge component requirements

5.13.1 MAC Bridge component options

Insert new subclause 5.13.1.1 as shown, renumbering as necessary:

5.13.1.1 Per-stream filtering and policing (PSFP) requirements

A MAC Bridge component implementation that conforms to the provisions of this standard for PSFP shall:

- a) Support PSFP as specified in 8.6.5.1 and 8.6.6.1.
- b) Support the state machines for stream gate control as specified in 8.6.10.
- c) Support the management entities for PSFP as specified in 12.31.

Insert the following new subclause at the end of Clause 5, renumbering as necessary:

5.27 End-station requirements—PSFP

An end-station implementation that conforms to the provisions of this standard for PSFP shall:

- a) Support PSFP as specified in 8.6.5.1 and 8.6.6.1.
- b) Support the state machines for stream gate control as specified in 8.6.10.
- c) Support the management entities for PSFP as specified in 12.31.

8. Principles of bridge operation

8.6 The Forwarding Process

Replace Figure 8-11 with the following figure. The effect of this replacement is to change “Ingress (8.6.2)” to “Ingress filtering (8.6.2)” and “Egress (8.6.4)” to “Egress filtering (8.6.4)” to match the subclause headings.

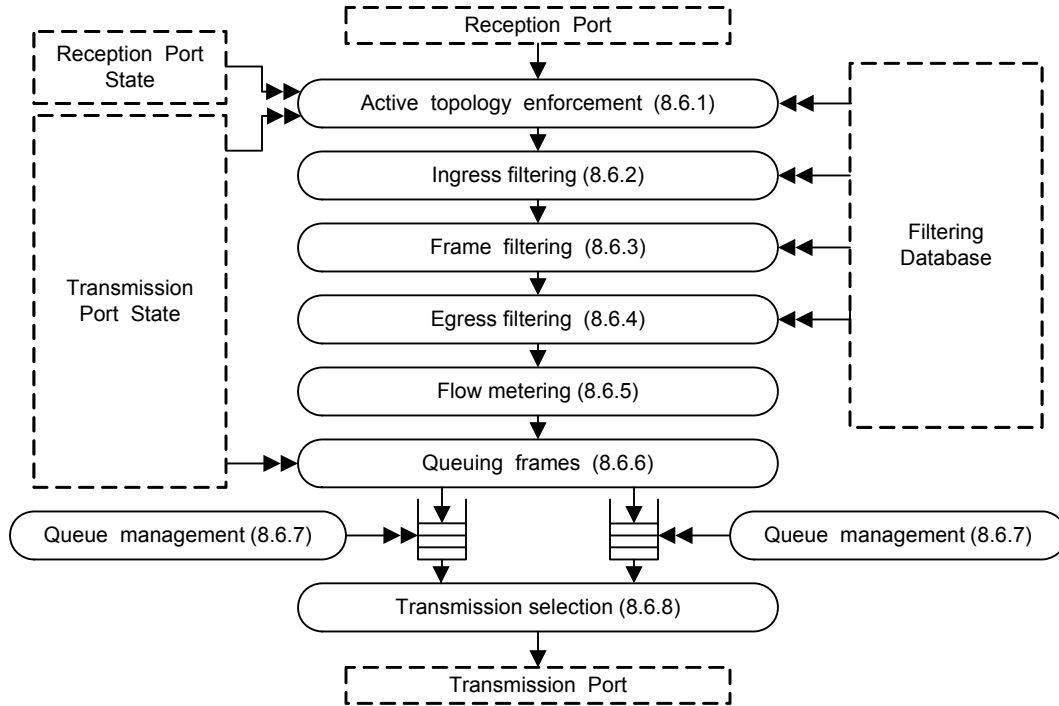


Figure 8-11—Forwarding process functions

8.6.5 Flow classification and metering

Insert new list item e) immediately following list item d), and change the sentence immediately following new item e), renumbering following list items, as shown:

- e) connection_identifier

Item c), specifying a VID value, is not applicable to VLAN-unaware MAC Relays. Item e), specifying a connection_identifier, is only applicable to bridges that support PSFP.

Insert new subclause 8.6.5.1, Figure 8-12, and Table 8-5 as shown, renumbering subsequent figures and tables as necessary:

8.6.5.1 Per-stream filtering and policing

A Bridge or an end station may support PSFP capabilities that allow filtering and policing decisions, and subsequent frame queuing decisions (8.6.6.1), to be made on a per-stream basis for received frames.

Support of PSFP requires implementation of the Stream identification function specified in Clause 6 of IEEE Std 802.1CB, as the stream_handle provided by this function is used in the policing and queuing decisions taken by PSFP.

NOTE—The *stream_handle* specified in Clause 6 of IEEE Std 802.1CB is a sub-parameter of the *connection_identifier* parameter of the ISS.

PSFP is supported by the following tables:

- a) Stream filter instance table (8.6.5.1.1)
- b) Stream gate instance table (8.6.5.1.2)
- c) Flow meter instance table (8.6.5.1.3)

The relationship between these tables is illustrated in Figure 8-12. The tables and their parameters can be modified by management as specified in 12.31.

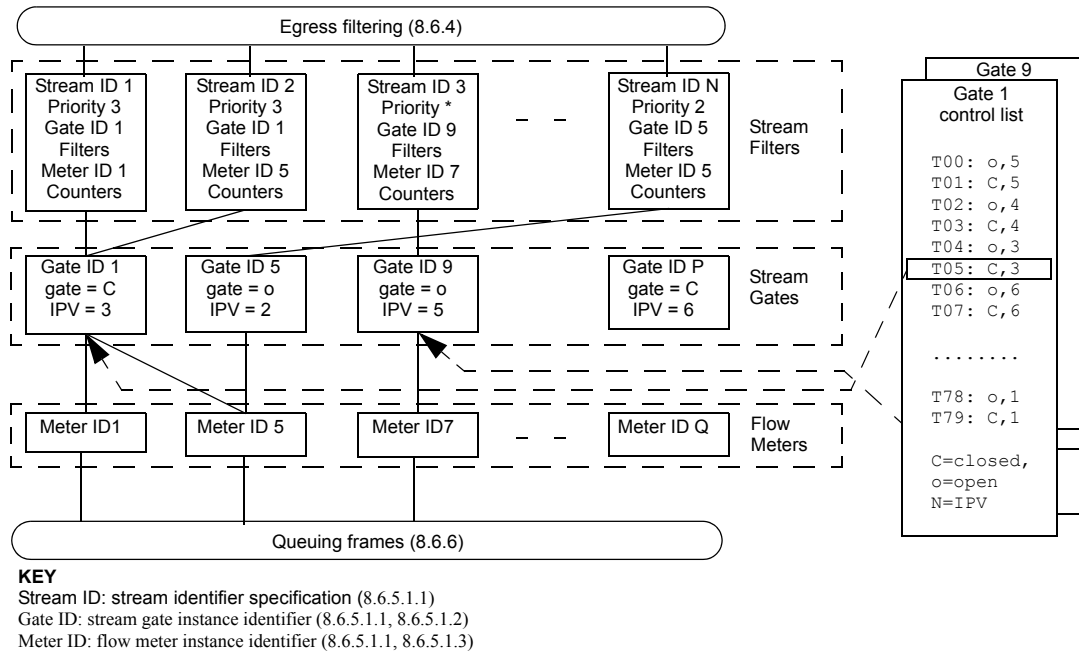


Figure 8-12—Per-stream filtering and policing

8.6.5.1.1 Stream filter instance table

The stream filter instance table consists of an ordered list of *stream filters* that determine the filtering and policing actions that are to be applied to frames received on a specific stream. Each stream filter contains the following elements:

- a) A *stream filter instance identifier*. This is an integer value that uniquely identifies the filter instance, and acts as an index to the table. The ordering of the identifier values defines the ordering of the list of stream filters; smaller identifier values appear earlier in the ordered list.
- b) A *stream_handle specification*. This can be either of the following:
 - 1) A single *stream_handle* value, as specified in IEEE Std 802.1CB.
 - 2) A wild-card value that matches any *stream_handle* value.
- c) A *priority specification*. This can be either of the following:
 - 1) A single priority value.
 - 2) A wild-card value that matches any priority value.

- d) A *stream gate instance identifier*. Identifies the stream gate instance (8.6.5.1.2) that is used by the stream filter. A stream gate can be in one of two states:
 - 1) Open: Frames pass through the gate.
 - 2) Closed: Frames do not pass through the gate.
- e) Zero or more *filter specifications*. The actions specified in a filter specification can result in a frame passing or failing the specified filter. Frames that fail a filter are discarded. The filter specification can include other actions, such as setting the `drop_eligible` parameter to TRUE. The following filter specifications are currently defined:
 - 1) *Maximum SDU size*. Frames that exceed this SDU size do not pass the stream filter; frames that do not exceed this SDU size can pass the stream filter if all other filter conditions are met.

NOTE 1—The Maximum SDU size is defined per stream and can therefore differ from the `queueMaxSDU` specified in 8.6.8.4 of IEEE Std 802.1Qbv. As `queueMaxSDU` is applied after the stream filters, it is possible that a frame that passes the Maximum SDU size stream filter will later be discarded because its SDU size exceeds `queueMaxSDU`.

- 2) *Flow meter instance identifier*. The identifier of an instance of a flow metering function as specified in 8.6.5. The flow meter instance is an index into a *flow meter instance table* (8.6.5.1.3) that specifies the operating parameters for each flow meter instance. Flow metering is always applied after any other filter specifications that could result in frame discard.
- f) Frame counters
 - 1) A count of frames matching both the `stream_handle` and priority specifications.
 - 2) A count of frames that passed the stream gate.
 - 3) A count of frames that did not pass the stream gate.
 - 4) A count of frames that passed the Maximum SDU size filter.
 - 5) A count of frames that did not pass the Maximum SDU size filter.
 - 6) A count of frames that were discarded as a result of the operation of the flow meter.
- g) A *StreamBlockedDueToOversizeFrameEnable* parameter, which takes the value TRUE or FALSE. A value of TRUE indicates that the *StreamBlockedDueToOversizeFrame* function is enabled; a value of FALSE indicates that the *StreamBlockedDueToOversizeFrame* function is disabled. The default value of *StreamBlockedDueToOversizeFrameEnable* is FALSE.
- h) A *StreamBlockedDueToOversizeFrame* parameter, which takes the value TRUE or FALSE. If *StreamBlockedDueToOversizeFrameEnable* is TRUE, a value of TRUE in *StreamBlockedDueToOversizeFrame* indicates that all frames are to be dropped (i.e., the stream filter behaves as it would if the maximum SDU size were to be set to 0 octets). If *StreamBlockedDueToOversizeFrame* is FALSE, it has no effect. The default value of *StreamBlockedDueToOversizeFrame* is FALSE; if any frame is discarded because it exceeds the Maximum SDU size for the stream, then *StreamBlockedDueToOversizeFrame* is set TRUE.

The value of the *stream_handle* and *priority* parameters associated with a received frame determine which stream filter is selected by the frame, and therefore what combination of filtering and policing actions is applied to the frame. If the *stream_handle* and *priority* parameters associated with a received frame match more than one stream filter, the stream filter that is selected is the one that appears earliest in the ordered list. If a received frame's *stream_handle* and *priority* does not match any of the stream filters in the table, the frame is processed as would be the case if PSFP was not supported.

NOTE 2—The use of stream identifier and priority, along with the wild-carding rules previously stated, allow configuration possibilities that go beyond PSFP as implied by the subclause title; for example, per-priority filtering and policing, or per-priority per-ingress port filtering and policing can be configured using these rules.

NOTE 3—If it is desired to discard frames that do not match any other stream filter, rather than such frames being processed without filtering, this can be achieved by placing a stream filter at the end of the table, in which the *stream_handle* and *priority* are both wild-carded (set to the null value), and where the stream gate instance identifier points at a stream gate that is permanently closed.

8.6.5.1.2 Stream gate instance table

The stream gate instance table contains a set of parameters for each stream gate instance. The parameters for each stream gate instance are as follows:

- a) A *Stream gate instance identifier*. An integer value identifying the stream gate instance.
- b) An operational and an administrative *stream gate state* (8.6.10.4, 8.6.10.5, 12.31.3). A stream gate can be in one of two states:
 - 1) Open: Frames are permitted to pass through the gate.
 - 2) Closed: Frames are not permitted to pass through the gate.
- c) An operational and an administrative *internal priority value specification (IPV)*, 8.6.10.6, 8.6.10.7, 12.31.3). An IPV can be either of the following:
 - 1) The null value. For a frame that passes through the gate, the priority value associated with the frame is used to determine the frame's traffic class, using the Traffic Class Table as specified in 8.6.6.
 - 2) An internal priority value. For a frame that passes through the gate, the IPV is used, in place of the priority value associated with the frame, to determine the frame's traffic class, using the Traffic Class Table as specified in 8.6.6.

NOTE 1—A use case for the ability to assign internal priority values can be found in IEEE Std 802.1Qch™ [B1].

- d) A *GateClosedDueToInvalidRxEnable* parameter, which takes the value TRUE or FALSE. A value of TRUE indicates that the GateClosedDueToInvalidRx function is enabled; a value of FALSE indicates that the GateClosedDueToInvalidRx function is disabled. The default value of GateClosedDueToInvalidRxEnable is FALSE.
- e) A *GateClosedDueToInvalidRx* parameter, which takes the value TRUE or FALSE. If GateClosedDueToInvalidRxEnable is TRUE, a value of TRUE in GateClosedDueToInvalidRx indicates that all frames are dropped (i.e., the gate behaves as if the operational stream gate state is Closed). If GateClosedDueToInvalidRx is FALSE, it has no effect. The default value of GateClosedDueToInvalidRx is FALSE; if any frame is discarded because the gate is in the Closed state, then GateClosedDueToInvalidRx is set TRUE.

NOTE 2—This parameter, in combination with its enable parameter, allow the detection of incoming frames during time periods when the stream gate is in the closed state to result in the stream gate being permanently set to a closed state, until such a time as management action is taken to reset the condition. The intent is to support applications where the transmission and reception of frames across the network is coordinated such that frames are received only when the stream gate is open, and hence, a frame received by the stream gate when it is in the closed state represents an invalid receive condition.

- f) A *GateClosedDueToOctetsExceededEnable* parameter, which takes the value TRUE or FALSE. A value of TRUE indicates that the GateClosedDueToOctetsExceeded function is enabled; a value of FALSE indicates that the GateClosedDueToOctetsExceeded function is disabled. The default value of GateClosedDueToOctetsExceededEnable is FALSE.
- g) A *GateClosedDueToOctetsExceeded* parameter, which takes the value TRUE or FALSE. If GateClosedDueToOctetsExceededEnable is TRUE, a value of TRUE in GateClosedDueToOctetsExceeded indicates that all frames are dropped (i.e., the gate behaves as if the operational stream gate state is Closed). If GateClosedDueToOctetsExceeded is FALSE, it has no effect. The default value of GateClosedDueToOctetsExceeded is FALSE; if any frame is discarded because there are insufficient IntervalOctetsLeft (8.6.10.8), then GateClosedDueToOctetsExceeded is set TRUE.
- h) Optionally, an operational and an administrative *stream gate control list*. If present, these are ordered lists of stream control operations, as specified in Table 8-7. The state machines that control the execution of the operational stream gate control list, along with their variables and procedures, are specified in 8.6.10.

Table 8-7—Stream gate control operations

Operation name	Parameter(s)	Action
SetGateAndIPV	StreamGateState, IPV, TimeInterval, IntervalOctetMax	The StreamGateState parameter specifies a desired state, <i>open</i> or <i>closed</i> , for the stream gate, and the IPV parameter specifies a desired value of the IPV associated with the stream. On execution, the StreamGateState and IPV parameter values are used to set the operational values of the stream gate state and internal priority specification parameters for the stream. After <i>TimeInterval</i> ticks (8.6.9.4.16) has elapsed since the completion of the previous stream gate control operation in the stream gate control list, control passes to the next stream gate control operation. The optional IntervalOctetMax parameter specifies the maximum number of MSDU octets that are permitted to pass the gate during the specified TimeInterval. If the IntervalOctetMax parameter is omitted, there is no limit on the number of octets that can pass the gate.

The administrative values of these parameters are used to determine the initial values of the corresponding operational parameters, and in the case of the administrative stream gate control list parameter, to provide a means of configuring a new control list prior to its installation in a running system.

8.6.5.1.3 Flow meter instance table

The flow meter instance table contains a set of parameters for each flow meter instance. The parameters for each flow meter instance are as specified in *Bandwidth Profile Parameters and Algorithm* in MEF 10.3, plus some additional parameters, as follows:

NOTE—Envelope and Rank, as defined in MEF 10.3, are not used for PSFP; i.e., PSFP uses the reduced functionality algorithm described in 12.2 of MEF 10.3.

- a) *Flow meter instance identifier*. An integer value identifying the flow meter instance.
- b) *Committed information rate (CIR)*, in bits per second.
- c) *Committed burst size (CBS)*, in octets.
- d) *Excess Information Rate (EIR)*, in bits per second.
- e) *Excess burst size (EBS) per bandwidth profile flow*, in octets.
- f) *Coupling flag (CF)*, which takes the value 0 or 1.
- g) *Color mode (CM)*, which takes the value *color-blind* or *color-aware*.
- h) *DropOnYellow*, which takes the value TRUE or FALSE. A value of TRUE indicates that yellow frames are dropped (i.e., discarded); a value of FALSE indicates that yellow frames will have the drop_eligible parameter set to TRUE.
- i) *MarkAllFramesRedEnable*, which takes the value TRUE or FALSE. A value of TRUE indicates that the MarkAllFramesRed function is enabled; a value of FALSE indicates that the MarkAllFramesRed function is disabled. The default value of MarkAllFramesRedEnable is FALSE.
- j) *MarkAllFramesRed*, which takes the value TRUE or FALSE. If MarkAllFramesRedEnable is TRUE, a value of TRUE in MarkAllFramesRed indicates that all frames are dropped (i.e., discarded). If MarkAllFramesRed is False, it has no effect. The default value of MarkAllFramesRed is FALSE; if the operation of the flow meter causes any frame to be discarded, then MarkAllFramesRed is set TRUE.

8.6.6 Queuing frames

Insert new subclause 8.6.6.1 as shown:

8.6.6.1 PSFP queuing

If PSFP is supported (8.6.5.1), and the IPV associated with the stream filter that passed the frame is anything other than the null value, then that IPV is used to determine the traffic class of the frame, in place of the frame's priority, via the Traffic Class Table specified in 8.6.6. In all other respects, the queuing actions specified in 8.6.6 are unchanged.

The IPV is used only to determine the traffic class associated with a frame, and hence select an outbound queue; for all other purposes, the received priority is used.

Insert new subclause 8.6.10, its subclauses, and Table 8-8, as shown, renumbering as necessary:

8.6.10 Stream gate control state machines

The execution of the gate operations in a stream gate control list (8.6.5.1.2) is controlled by the three state machines specified in 8.6.9³:

- a) The Cycle Timer state machine (8.6.9.1);
- b) The List Execute state machine (8.6.9.2); and
- c) The List Config state machine (8.6.9.3).

One instance of each state machine is instantiated for each stream gate control list associated with instances of stream gates in a Bridge component that supports PSFP. An overview of the operation of these state machines can be found in Figure 8-13.

The operation of these state machines is as defined in 8.6.9, with the exception of the definitions of the ExecuteOperation() procedure, the SetGateStates() procedure, the ListPointer variable, the AdminGateStates variable, and the OperGateStates variable; amended versions of these definitions appear in 8.6.10.1 through 8.6.10.5. Table 8-8 shows the correspondence between the procedures/variables used in 8.6.9 and the PSFP versions of these procedures/variables.

Three additional variables needed by the ExecutePSFPOperation procedure are defined in 8.6.10.6 and 8.6.10.7.

Table 8-8—Scheduled Traffic and PSFP procedures/variables

Procedure/variable name in 8.6.9	PSFP procedure/variable name
ExecuteOperation() (8.6.9.2.1)	ExecutePSFPOperation() (8.6.10.1)
SetGateStates() (8.6.9.2.2)	SetPSFPGateStates() (8.6.10.2)
ListPointer (8.6.9.4.15)	PSFPListPointer (8.6.10.3)
AdminGateStates (8.6.9.4.5)	PSFPAdminGateStates (8.6.10.4)
OperGateStates (8.6.9.4.22)	PSFPOperGateStates (8.6.10.5)

³Figure 8-13 and 8.6.9 can be found IEEE Std 802.1Qbv, which is also an amendment to IEEE Std 802.1Q.

8.6.10.1 ExecutePSFPOperation()

The ExecutePSFPOperation() procedure is responsible for fetching the next gate operation from the OperControlList, along with any parameters associated with it, and performing actions based upon the gate operation that has been fetched. The value of the PSFPListPointer variable (8.6.10.3) is used as an index into OperControlList. The procedure processes the operation according to its operation name (Table 8-7) as follows:

- a) If the operation name is SetGateAndIPV, then the StreamGateState parameter value associated with the operation is assigned to the PSFPOperGateStates variable (8.6.10.5), the IPV parameter value is assigned to the OperIPV variable (8.6.10.7), and the TimeInterval parameter value associated with the operation is assigned to the TimeInterval variable (8.6.9.4.24). If the TimeInterval parameter value associated with the operation was 0, the TimeInterval variable is assigned the value 1. If there is an IntervalOctetMax parameter associated with the gate operation, then that parameter value is used to set the value of the IntervalOctetsLeft variable (8.6.10.8); otherwise, the IntervalOctetsLeft variable is set to a value greater than the maximum possible number of octets that the gate could pass during TimeInterval.
- b) If the operation name is unrecognized, then the PSFPListPointer variable (8.6.9.4.15) is assigned the value of the OperControlListLength variable (8.6.9.4.23) and the TimeInterval variable (8.6.9.4.24) is assigned the value 0.
- c) If there is no TimeInterval parameter associated with the operation, then the TimeInterval variable is assigned the value 0.

8.6.10.2 SetPSFPGateStates()

This procedure sets the stream gate state as specified by the value of the PSFPOperGateStates variable (8.6.9.4.22).

8.6.10.3 PSFPListPointer

An integer used as a pointer to entries in the OperControlList (8.6.9.4.19), each entry consisting of a stream gate control operation with its associated parameters (Table 8-7). A value of zero points at the first entry in the list; a value of (OperControlListLength) – 1 points at the last entry.

8.6.10.4 PSFPAdminGateStates

The initial state of the gate associated with the stream gate is set by the List Execute state machine (8.6.9.2) and is determined by the value of the PSFPAdminGateStates variable. The default value of PSFPAdminGateStates is open. The value of PSFPAdminGateStates can be changed by management.

8.6.10.5 PSFPOperGateStates

The current state of the gate associated with the stream gate. PSFPOperGateStates is set by the List Execute state machine (8.6.9.2), and its initial value is determined by the value of the PSFPAdminGateStates variable (8.6.10.4).

8.6.10.6 AdminIPV

The initial value of the OperIPV variable (8.6.10.7) associated with the stream gate is determined by the value of the AdminIPV variable. The default value of AdminIPV variable is the null value. The value of the AdminIPV variable can be changed by management.

8.6.10.7 OperIPV

The current value of the IPV associated with the stream gate. The initial value of OperIPV is set equal to the value of the AdminIPV variable (8.6.10.6). Subsequently, if there is a stream gate control list associated with the stream gate instance, its value is controlled by the contents of the operational stream gate control list and the operation of the List Execute state machine (8.6.9.2).

8.6.10.8 IntervalOctetsLeft

The current value of the IntervalOctetsLeft parameter indicates how many more MSDU octets can be passed by the stream gate during the current TimeInterval. This variable is initialized by the ExecutePSFPOperation() procedure (8.6.10.1). If a frame that would otherwise pass the gate is larger than the current value of IntervalOctetsLeft, it is treated as if the gate is in the *closed* state; i.e., it is discarded. If a frame that would otherwise pass the gate is smaller than the current value of IntervalOctetsLeft, the number of MSDU octets is subtracted from the value of IntervalOctetsLeft.

12. Bridge management

Insert new subclause 12.31, and its subclauses and tables, as shown, renumbering as necessary.

12.31 Managed objects for per-stream filtering and policing

The Bridge enhancements for support of per-stream filtering and policing are defined in 8.6.5.1 and the associated state machines are defined in 8.6.10.

The objects that comprise this managed resource are as follows:

- a) The Stream Parameter Table (12.31.1)
- b) The Stream Filter Instance Table (12.31.2)
- c) The Stream Gate Instance Table (12.31.3)
- d) The Flow Meter Instance Table (12.31.4)

12.31.1 The Stream Parameter Table

There is one Stream Parameter Table per Bridge component. The table contains a set of parameters that supports PSFP (8.6.5.1), as detailed in Table 12-30. Tables can be created or removed dynamically in implementations that support dynamic configuration of Bridge components.

Table 12-30—The Stream Parameter Table

Name	Data type	Operations supported ^a	Conformance ^b	References
MaxStreamFilterInstances	integer	R	BE	8.6.5.1, 12.31.2
MaxStreamGateInstances	integer	R	BE	8.6.5.1, 12.31.3
MaxFlowMeterInstances	integer	R	BE	8.6.5.1, 12.31.4
SupportedListMax	integer	R	BE	8.6.5.1, 12.31.4

^aR= Read only access; RW = Read/Write access.

^bB = Required for Bridge or Bridge component support of PSFP.

E = Required for end-station support of PSFP.

12.31.1.1 MaxStreamFilterInstances

The maximum number of Stream Filter instances supported by this Bridge component.

12.31.1.2 MaxStreamGateInstances

The maximum number of Stream Gate instances supported by this Bridge component.

12.31.1.3 MaxFlowMeterInstances

The maximum number of Flow Meter instances supported by this Bridge component.

12.31.1.4 SupportedListMax

The maximum value supported by this Bridge component of the AdminControlListLength and OperControlListLength parameters. It is available for use by schedule computation software to determine the Bridge component's control list capacity prior to computation.

12.31.2 The Stream Filter Instance Table

There is one Stream Filter Instance Table per Bridge component. Each table row contains a set of parameters that defines a single Stream Filter (8.6.5.1), as detailed in Table 12-31. The table rows form an ordered list of filter instances, the order being determined by the StreamFilterInstance parameter. Tables can be created or removed dynamically in implementations that support dynamic configuration of Bridge components. Rows in the table can be created or removed dynamically in implementations that support dynamic configuration of stream filters.

Table 12-31—Stream Filter Instance Table

Name	Data type	Operations supported ^a	Conformance ^b	References
StreamFilterInstance	integer	R	BE	8.6.5.1
StreamHandleSpec	stream_handle specification	RW	BE	8.6.5.1
PrioritySpec	priority specification	RW	BE	8.6.5.1
StreamGateInstanceID	integer	RW	BE	8.6.5.1, 8.6.5.1.2
FilterSpecificationList	sequence of FilterSpecification	RW	BE	8.6.5.1, 8.6.5.1.3, 12.31.2.5
MatchingFramesCount	counter	R	BE	8.6.5.1
PassingFramesCount	counter	R	BE	8.6.5.1
NotPassingFramesCount	counter	R	BE	8.6.5.1
PassingSDUCount	counter	R	BE	8.6.5.1
NotPassingSDUCount	counter	R	BE	8.6.5.1
REDFramesCount	counter	R	BE	8.6.5.1
StreamBlockedDueToOver-sizeFrameEnable	Boolean	RW	BE	8.6.5.1, 8.6.5.1.1
StreamBlockedDueToOver-sizeFrame	Boolean	RW	BE	8.6.5.1, 8.6.5.1.1

^aR= Read only access; RW = Read/Write access.

^bB = Required for Bridge or Bridge component support of PSFP.

E = Required for end-station support of PSFP.

12.31.2.1 StreamFilterInstance

An integer index value that determines the place of the stream filter in the ordered list of stream filter instances. The values of StreamFilterInstance are ordered according to their integer value; smaller values appear earlier in the ordered list.

12.31.2.2 stream_handle specification data type

The stream_handle specification data type allows either of the following to be represented:

- a) A stream_handle value, represented as an integer.
- b) The wild card value.

12.31.2.3 priority specification data type

The priority specification data type allows either of the following to be represented:

- a) A priority value, represented as an integer.
- b) The wild card value.

12.31.2.4 StreamGateInstance

The StreamGateInstance parameter identifies the stream gate (12.31.3) that is associated with the stream filter. The relationship between stream filters and stream gates is many to one; a given stream filter can be associated with only one stream gate, but there can be multiple stream filters associated with a given stream gate.

12.31.2.5 FilterSpecification data type

The FilterSpecification data type can represent the following:

- a) An integer value representing a Maximum SDU size (8.6.5.1).
- b) An integer value representing a flow meter instance identifier (8.6.5.1, 8.6.5.1.3).

12.31.3 The Stream Gate Instance Table

There is one Stream Gate Instance Table per Bridge component. Each table row contains a set of parameters that defines a single Stream Gate (8.6.5.1.2), as detailed in Table 12-32. Tables can be created or removed dynamically in implementations that support dynamic configuration of Bridge components. Rows in the table can be created or removed dynamically in implementations that support dynamic configuration of stream gates.

Table 12-32—The Stream Gate Instance Table

Name	Data type	Operations supported ^a	Conformance ^b	References
StreamGateInstance	integer	R	BE	8.6.5.1, 8.6.5.1.2
PSFPGateEnabled	Boolean	RW	BE	8.6.9.4.14
PSFPAdminGateStates	PSFPgateStatesValue	RW	BE	8.6.10.4, 12.31.3.2.1
PSFPOperGateStates	PSFPgateStatesValue	R	BE	8.6.10.5, 12.31.3.2.1
PSFPAdminControlListLength	unsigned integer	RW	BE	8.6.9.4.6, 12.31.3.2
PSFPOperControlListLength	unsigned integer	R	BE	8.6.9.4.23, 12.31.3.2
PSFPAdminControlList	sequence of PSFP- GateControlEntry	RW	BE	8.6.9.4.2, 12.31.3.2, 12.31.3.2.2
PSFPOperControlList	sequence of PSFP- GateControlEntry	R	BE	8.6.9.4.19, 12.31.3.2, 12.31.3.2.2
PSFPAdminCycleTime	RationalNumber	RW	BE	8.6.9.4.3, 12.29.1.3
PSFPOperCycleTime	RationalNumber (seconds)	R	BE	8.6.9.4.20, 12.29.1.3
PSFPAdminCycleTimeExtension	Integer (nanoseconds)	RW	BE	8.6.9.4.4
PSFPOperCycleTimeExtension	Integer (nanoseconds)	R	BE	8.6.9.4.21
PSFPAdminBaseTime	PTPtime	RW	BE	8.6.9.4.1, 12.29.1.4
PSFPOperBaseTime	PTPtime	R	BE	8.6.9.4.18, 12.29.1.4
PSFPConfigChange	Boolean	RW	BE	8.6.9.4.7
PSFPConfigChangeTime	PTPtime	R	BE	8.6.9.4.9, 12.29.1.4
PSFPTickGranularity	Integer (tenths of nanoseconds)	R	BE	8.6.9.4.16
PSFPCurrentTime	PTPtime	R	BE	8.6.9.4.10, 12.29.1.4
PSFPConfigPending	Boolean	R	BE	8.6.9.3, 8.6.9.4.8
PSFPConfigChangeError	Integer	R	BE	8.6.9.3.1

Table 12-32—The Stream Gate Instance Table (*continued*)

Name	Data type	Operations supported ^a	Conformance ^b	References
PSFPAdminIPV	IPV	RW	BE	8.6.5.1.2, 8.6.10.6, 12.31.3.3
PSFPOperIPV	IPV	RW	BE	8.6.5.1.2, 8.6.10.7, 12.31.3.3
PSFPGateClosedDueToInvalidRx-Enable	Boolean	RW	BE	8.6.5.1.2
PSFPGateClosedDueToInvalidRx	Boolean	RW	BE	8.6.5.1.2
PSFPGateClosedDueToOctets-ExceededEnable	Boolean	RW	BE	8.6.5.1.2
PSFPGateClosedDueToOctetsExceeded	Boolean	RW	BE	8.6.5.1.2

^aR= Read only access; RW = Read/Write access.

^bB = Required for Bridge or Bridge component support of PSFP.

E = Required for end-station support of PSFP.

12.31.3.1 StreamGateInstance

An integer table index that allows the stream gate to be referenced from Stream Filter Instance Table entries.

12.31.3.2 The gate control list structure and data types

The AdminControlList and OperControlList are ordered lists containing AdminControlListLength or OperControlListLength entries, respectively. Each entry represents a gate operation as defined in Table 8-7. Each entry in the list is structured as a GateControlEntry (12.31.3.2.2).

12.31.3.2.1 PSFPgateStatesValue

The PSFPgateStatesValue indicates the desired gate state, *open* or *closed*, for the stream gate.

12.31.3.2.2 PSFPGateControlEntry

A PSFPGateControlEntry consists of an operation name, followed by three parameters associated with the operation, as detailed in Table 8-7. The first parameter is a PSFPgateStatesValue (12.31.3.2.1); the second parameter is an IPV value (12.31.3.2.3), and the third parameter is a timeIntervalValue (12.31.3.2.4).

12.31.3.2.3 IPV value

The IPV value indicates the IPV (12.31.3.3) to be associated with frames that pass the gate (8.6.10.7).

12.31.3.2.4 timeIntervalValue

An unsigned integer, denoting a TimeInterval in nanoseconds (see TimeInterval in Table 8-7).

12.31.3.3 The Internal priority value specification (IPV) data type

The IPV data type represents an IPV value (8.6.5.1.2); this is either the null value or an internal priority value.

12.31.4 The Flow Meter Instance Table

There is one Flow Meter Instance Table per Bridge component. Each table row contains a set of parameters that defines a single Flow Meter Instance (8.6.5.1), as detailed in Table 12-33. Tables can be created or removed dynamically in implementations that support dynamic configuration of Bridge components. Rows in the table can be created or removed dynamically in implementations that support dynamic configuration of flow meters.

Table 12-33—The Flow Meter Instance Table

Name	Data type	Operations supported ^a	Conformance ^b	References
FlowMeterInstanceID	integer	R	BE	8.6.5.1, 8.6.5.1.3
CIR	integer, bit/s	RW	BE	8.6.5.1, 8.6.5.1.3
CBS	integer, octets	RW	BE	8.6.5.1, 8.6.5.1.3
EIR	integer, bit/s	RW	BE	8.6.5.1, 8.6.5.1.3
EBS	integer, octets	RW	BE	8.6.5.1, 8.6.5.1.3
CF	integer, 0 or 1	RW	BE	8.6.5.1, 8.6.5.1.3
CM	enumerated, color-blind or color-aware	RW	BE	8.6.5.1, 8.6.5.1.3
DropOnYellow	Boolean	RW	BE	8.6.5.1, 8.6.5.1.3
MarkAllFramesRedEnable	Boolean	RW	BE	8.6.5.1, 8.6.5.1.3
MarkAllFramesRed	Boolean	RW	BE	8.6.5.1, 8.6.5.1.3

^aR= Read only access; RW = Read/Write access.

^bB = Required for Bridge or Bridge component support of PSFP.

E = Required for end-station support of PSFP.

17. Management Information Base (MIB)

17.2 Structure of the MIB

Insert new subclause 17.2.24 and Table 17-30 at the end of 17.2, as shown, renumbering as necessary.

17.2.24 Structure of the IEEE8021-PSFP-MIB

The IEEE8021-PSFP-MIB provides for configuration of PSFP (8.6.5, 8.6.5.1, 8.6.10) on reception Ports. Table 17-30 indicates the relationship between the SMIPv2 objects defined in the MIB module (17.7.24) and managed objects defined in 12.31.

Table 17-30—IEEE8021-PSFP-MIB Structure and relationship to this standard

MIB table	MIB object	Reference
<i>ieee8021PSFPStreamFilterParameters subtree</i>		
	ieee8021PSFPStreamFilterTable	Stream Filter Instance Table, 8.6.5, 8.6.5.1, 12.31.2
	ieee8021PSFPStreamFilterInstance	StreamFilterInstance, 8.6.5, 8.6.5.1, 12.31.2
	ieee8021PSFPStreamHandleSpec	StreamHandleSpec, 8.6.5, 8.6.5.1, 12.31.2
	ieee8021PSFPPrioritySpec	PrioritySpec, 8.6.5, 8.6.5.1, 12.31.2
	ieee8021PSFPStreamGateInstanceID	StreamGateInstanceID, 8.6.5, 8.6.5.1, 12.31.2
	ieee8021PSFPFilterSpecificationList	FilterSpecificationList, 8.6.5, 8.6.5.1, 12.31.2
	ieee8021PSFPMatchingFramesCount	MatchingFramesCount, 8.6.5, 8.6.5.1, 12.31.2
	ieee8021PSFPPassingFramesCount	PassingFramesCount, 8.6.5, 8.6.5.1, 12.31.2
	ieee8021PSFPNotPassingFramesCount	NotPassingFramesCount, 8.6.5, 8.6.5.1, 12.31.2
	ieee8021PSFPPassingSDUCount	PassingSDUCount, 8.6.5, 8.6.5.1, 12.31.2
	ieee8021PSFPNotPassingSDUCount	NotPassingSDUCount, 8.6.5, 8.6.5.1, 12.31.2
	ieee8021PSFPPREDFramesCount	REDFramesCount, 8.6.5, 8.6.5.1, 12.31.2
	ieee8021PSFPStreamBlockedDueToOversizeFrameEnable	StreamBlockedDueToOversizeFrameEnable, 8.6.5.1, 8.6.5.1.1, 12.31.2
	ieee8021PSFPStreamBlockedDueToOversizeFrame	StreamBlockedDueToOversizeFrame, 8.6.5.1, 8.6.5.1.1, 12.31.2

Table 17-30—IEEE8021-PSFP-MIB Structure and relationship to this standard (continued)

MIB table	MIB object	Reference
<i>ieee8021PSFPStreamGateParameters</i>		
ieee8021PSFPStreamGateTable		Stream Gate Instance Table, 8.6.5, 8.6.5.1, 12.31.3
	ieee8021PSFPStreamGateInstance	StreamGateInstance, 8.6.5, 8.6.5.1, 12.31.3
	ieee8021PSFPGateEnabled	PSFPGateEnabled, 8.6.5, 8.6.5.1, 8.6.10, 12.31.3
	ieee8021PSFPAdminGateStates	PSFPAdminGateStates, 8.6.5, 8.6.5.1, 8.6.10, 12.31.3
	ieee8021PSFPOperGateStates	PSFPOperGateStates, 8.6.5, 8.6.5.1, 8.6.10, 12.31.3
	ieee8021PSFPAdminControlListLength	PSFPAdminControlListLength, 8.6.5, 8.6.5.1, 8.6.10, 12.31.3
	ieee8021PSFPOperControlListLength	PSFPOperControlListLength, 8.6.5, 8.6.10, 12.31.3
	ieee8021PSFPAdminControlList	PSFPAdminControlList, 8.6.5, 8.6.5.1, 8.6.10, 12.31.3
	ieee8021PSFPOperControlList	PSFPOperControlList, 8.6.5, 8.6.5.1, 8.6.10, 12.31.3
	ieee8021PSFPAdminCycleTimeNumerator	PSFPAdminCycleTime, 8.6.5, 8.6.5.1, 8.6.10, 12.31.3
	ieee8021PSFPAdminCycleTimeDenominator	PSFPAdminCycleTime, 8.6.5, 8.6.5.1, 8.6.10, 12.31.3
	ieee8021PSFPOperCycleTimeNumerator	PSFPOperCycleTime, 8.6.5, 8.6.5.1, 8.6.10, 12.31.3
	ieee8021PSFPOperCycleTimeDenominator	PSFPOperCycleTime, 8.6.5, 8.6.5.1, 8.6.10, 12.31.3
	ieee8021PSFPAdminCycleTimeExtension	PSFPAdminCycleTimeExtension, 8.6.5, 8.6.5.1, 8.6.10, 12.31.3
	ieee8021PSFPOperCycleTimeExtension	PSFPOperCycleTimeExtension, 8.6.5, 8.6.5.1, 8.6.10, 12.31.3
	ieee8021PSFPAdminBaseTime	PSFPAdminBaseTime, 8.6.5, 8.6.5.1, 8.6.10, 12.31.3
	ieee8021PSFPOperBaseTime	PSFPOperBaseTime, 8.6.5, 8.6.5.1, 8.6.10, 12.31.3
	ieee8021PSFPConfigChange	PSFPConfigChange, 8.6.5, 8.6.5.1, 8.6.10, 12.31.3
	ieee8021PSFPConfigChangeTime	PSFPConfigChangeTime, 8.6.5, 8.6.5.1, 8.6.10, 12.31.3
	ieee8021PSFPTickGranularity	PSFPTickGranularity, 8.6.5, 8.6.5.1, 8.6.10, 12.31.3

Table 17-30—IEEE8021-PSFP-MIB Structure and relationship to this standard (continued)

MIB table	MIB object	Reference
	ieee8021PSFPCurrentTime	PSFPCurrentTime, 8.6.5, 8.6.5.1, 8.6.10, 12.31.3
	ieee8021PSFPConfigPending	PSFPConfigPending, 8.6.5, 8.6.5.1, 8.6.10, 12.31.3
	ieee8021PSFPConfigChangeError	PSFPConfigChangeError, 8.6.5, 8.6.5.1, 8.6.10, 12.31.3
	ieee8021PSFPAdminIPV	PSFPAdminIPV, 8.6.5, 8.6.5.1, 12.31.3
	ieee8021PSFPOperIPV	PSFPOperIPV, 8.6.5, 8.6.5.1, 12.31.3
	ieee8021PSFPGateClosedDueToInvalidRxEnable	PSFPGateClosedDueToInvalidRx-Enable, 8.6.5.1.2
	ieee8021PSFPGateClosedDueToInvalidRx	PSFPGateClosedDueToInvalidRx, 8.6.5.1.2
	ieee8021PSFPGateClosedDueToOctetsExceededEnable	PSFPGateClosedDueToOctets-ExceededEnable, 8.6.5.1.2
	ieee8021PSFPGateClosedDueToOctetsExceeded	PSFPGateClosedDueToOctets-Exceeded, 8.6.5.1.2
<i>ieee8021PSFPFlowMeterParameters</i>		
	ieee8021PSFPFlowMeterTable	Flow Meter Instance Table, 8.6.5, 8.6.5.1, 12.31.4
	ieee8021PSFPFlowMeterInstance	FlowMeterInstanceID, 8.6.5, 8.6.5.1, 12.31.4
	ieee8021PSFPFlowMeterCIR	CIR, 8.6.5, 8.6.5.1, 12.31.4
	ieee8021PSFPFlowMeterCBS	CBS, 8.6.5, 8.6.5.1, 12.31.4
	ieee8021PSFPFlowMeterEIR	EIR, 8.6.5, 8.6.5.1, 12.31.4
	ieee8021PSFPFlowMeterCF	CF, 8.6.5, 8.6.5.1, 12.31.4
	ieee8021PSFPFlowMeterCM	CM, 8.6.5, 8.6.5.1, 12.31.4
	ieee8021PSFPFlowMeterDropOnYellow	DropOnYellow, 8.6.5, 8.6.5.1, 12.31.4
	ieee8021PSFPFlowMeterMarkAllFramesRedEnable	MarkAllFramesRedEnable, 8.6.5, 8.6.5.1, 12.31.4
	ieee8021PSFPFlowMeterMarkAllFramesRed	MarkAllFramesRed, 8.6.5, 8.6.5.1, 12.31.4

Table 17-30—IEEE8021-PSFP-MIB Structure and relationship to this standard (continued)

MIB table	MIB object	Reference
<i>ieee8021PSFPStreamParameters</i>		
ieee8021PSFPStreamParameterTable		StreamParameterTable, 8.6.5, 8.6.5.1, 12.31.1
	ieee8021PSFPMaxStreamFilterInstances	MaxStreamFilterInstances, 8.6.5, 8.6.5.1, 12.31.1
	ieee8021PSFPMaxStreamGateInstances	MaxStreamGateInstances, 8.6.5, 8.6.5.1, 12.31.1
	ieee8021PSFPMaxFlowMeterInstances	MaxFlowMeterInstances, 8.6.5, 8.6.5.1, 12.31.1
	ieee8021PSFPSupportedListMax	SupportedListMax, 8.6.5, 8.6.5.1, 12.31.1

17.3 Relationship to other MIBs

Insert new subclause 17.3.24 at the end of 17.3, as shown, renumbering as necessary.

17.3.24 Relationship of IEEE8021-PSFP-MIB to other MIBs

The IEEE8021-PSFP-MIB provides objects that extend the core management functionality of a Bridge, as defined by the IEEE8021-BRIDGE-MIB (17.7.2), in order to support the additional management functionality needed when the PSFP extensions, as defined in 8.6.5 and 8.6.10, are supported by the Bridge. As support of the objects defined in the IEEE8021-PSFP-MIB also requires support of the IEEE8021-BRIDGE-MIB, the provisions of 17.3.2 apply to implementations claiming support of the IEEE8021-PSFP-MIB.

17.4 Security considerations

Insert new subclause 17.4.24 at the end of 17.4, as shown, renumbering as necessary.

17.4.24 Security considerations of the IEEE8021-PSFP-MIB

There are a number of management objects defined in the IEEE8021-PSFP-MIB module that have a MAX-ACCESS clause of read-write or read-create. Such objects may be considered sensitive or vulnerable in some network environments. The support for SET operations in a nonsecure environment without proper protection can have a negative effect on network operations.

The following tables and objects in the IEEE8021-PSFP-MIB can be misconfigured to interfere with the operation of the forwarding and queuing mechanisms in a manner that would be detrimental to the operation of PSFP:

- ieee8021PSFPStreamHandleSpec
- ieee8021PSFPPrioritySpec
- ieee8021PSFPStreamGateInstanceID
- ieee8021PSFPFilterSpecificationList
- ieee8021PSFPStreamBlockedDueToOversizeFrameEnable
- ieee8021PSFPStreamBlockedDueToOversizeFrame

ieee8021PSFPGateEnabled
ieee8021PSFPAdminGateStates
ieee8021PSFPAdminControlListLength
ieee8021PSFPAdminControlList
ieee8021PSFPAdminCycleTimeNumerator
ieee8021PSFPAdminCycleTimeDenominator
ieee8021PSFPAdminCycleTimeExtension
ieee8021PSFPAdminBaseTime
ieee8021PSFPConfigChange
ieee8021PSFPConfigChangeTime
ieee8021PSFPAdminIPV
ieee8021PSFPOperIPV
ieee8021PSFPGateClosedDueToInvalidRxEnable
ieee8021PSFPGateClosedDueToInvalidRx
ieee8021PSFPGateClosedDueToOctetsExceededEnable
ieee8021PSFPGateClosedDueToOctetsExceeded
ieee8021PSFPFlowMeterInstance
ieee8021PSFPFlowMeterCIR
ieee8021PSFPFlowMeterCBS
ieee8021PSFPFlowMeterEIR
ieee8021PSFPFlowMeterCF
ieee8021PSFPFlowMeterCM
ieee8021PSFPFlowMeterDropOnYellow
ieee8021PSFPFlowMeterMarkAllFramesRedEnable
ieee8021PSFPFlowMeterMarkAllFramesRed

- a) ieee8021PSFPStreamHandleSpec, ieee8021PSFPPrioritySpec, ieee8021PSFPStreamGateInstanceID, and ieee8021PSFPFilterSpecificationList can be misconfigured to adversely affect the policing functions that are applied to received frames.
- b) ieee8021PSFPGateEnabled can be misconfigured to enable/disable scheduled traffic processing.
- c) ieee8021PSFPAdminGateStates can be misconfigured to affect the gate state of a stream filter on startup.
- d) ieee8021PSFPAdminControlListLength, ieee8021PSFPAdminControlList, ieee8021PSFPAdminCycleTimeNumerator, ieee8021PSFPAdminCycleTimeDenominator, ieee8021PSFPAdminCycleTimeExtension, ieee8021PSFPAdminBaseTime, ieee8021PSFPConfigChange, and ieee8021PSFPConfigChangeTime can be misconfigured to affect the filter schedule for the Port.
ieee8021PSFPFlowMeterInstance, ieee8021PSFPFlowMeterCIR, ieee8021PSFPFlowMeterCBS, ieee8021PSFPFlowMeterEIR, ieee8021PSFPFlowMeterCF, ieee8021PSFPFlowMeterCM, ieee8021PSFPFlowMeterDropOnYellow, ieee8021PSFPFlowMeterMarkAllFramesRedEnable, ieee8021PSFPFlowMeterMarkAllFramesRed, ieee8021PSFPGateClosedDueToInvalidRxEnable, ieee8021PSFPGateClosedDueToInvalidRx, ieee8021PSFPGateClosedDueToOctetsExceededEnable, ieee8021PSFPGateClosedDueToOctetsExceeded, ieee8021PSFPStreamBlockedDueToOversizeFrameEnable, and ieee8021PSFPStreamBlockedDueToOversizeFrame can be misconfigured to adversely affect the way that flow metering operates.

Some of the readable objects in this MIB module (i.e., objects with a MAX-ACCESS other than not accessible) can be considered sensitive or vulnerable in some network environments. It is thus important to control all types of access (including GET and/or NOTIFY) to these objects and possibly to encrypt the values of these objects when sending them over the network via SNMP.

17.7 MIB modules

Insert new subclause 17.7.24 at the end of 17.7, as shown, renumbering as necessary.

17.7.24 Definitions for the IEEE8021-PSFP-MIB module

```
IEEE8021-PSFP-MIB DEFINITIONS ::= BEGIN

-- =====
-- MIB for support of the Per-Stream Filtering and Policing
-- Enhancements for 802.1Q Bridges.
-- =====

IMPORTS
    MODULE-IDENTITY,
    OBJECT-TYPE,
    Unsigned32,
    Integer32,
    Counter64
        FROM SNMPv2-SMI
    TruthValue, RowStatus
        FROM SNMPv2-TC
    MODULE-COMPLIANCE,
    OBJECT-GROUP
        FROM SNMPv2-CONF
    ieee802dot1mibs
        FROM IEEE8021-TC-MIB
    ieee8021BridgeBaseComponentId
        FROM IEEE8021-BRIDGE-MIB
    IEEE8021STPTPtimeValue
        FROM IEEE8021-ST-MIB
    ;

ieee8021PSFPMib MODULE-IDENTITY
    LAST-UPDATED "201709080000Z" -- October 6, 2017
    ORGANIZATION "IEEE 802.1 Working Group"
    CONTACT-INFO
        " WG-URL: http://www.ieee802.org/1/
          WG-EMail: stds-802-1-L@ieee.org

          Contact: IEEE 802.1 Working Group Chair
          Postal: C/O IEEE 802.1 Working Group
                  IEEE Standards Association
                  445 Hoes Lane
                  P.O. Box 1331
                  Piscataway
                  NJ 08855-1331
                  USA
          E-mail: stds-802-1-L@ieee.org"
    DESCRIPTION
        "The Bridge MIB module for managing devices that support
        the Per-Stream Filtering and Policing enhancements
        for 802.1Q Bridges.

        Unless otherwise indicated, the references in this MIB
        module are to IEEE Std 802.1Q-2014.

        Copyright (C) IEEE (2017)."
```

IEEE Std 802.1Qci-2017
IEEE Standard for Local and Metropolitan Area Networks—Bridges and Bridged Networks—
Amendment 28: Per-Stream Filtering and Policing

This version of this MIB module is part of IEEE802.1Q;
see the draft itself for full legal notices."

REVISION "201709080000Z" -- October 6, 2017

DESCRIPTION

"Initial version published as part of IEEE Std 802.1Qci."

::= { ieee802dot1mibs 31 }

-- =====
-- subtrees in the PSFP MIB
-- =====

ieee8021PSFPNotifications

OBJECT IDENTIFIER ::= { ieee8021PSFPMib 0 }

ieee8021PSFPObjects

OBJECT IDENTIFIER ::= { ieee8021PSFPMib 1 }

ieee8021PSFPConformance

OBJECT IDENTIFIER ::= { ieee8021PSFPMib 2 }

ieee8021PSFPStreamFilterParameters

OBJECT IDENTIFIER ::= { ieee8021PSFPObjects 1 }

ieee8021PSFPStreamGateParameters

OBJECT IDENTIFIER ::= { ieee8021PSFPObjects 2 }

ieee8021PSFPFlowMeterParameters

OBJECT IDENTIFIER ::= { ieee8021PSFPObjects 3 }

ieee8021PSFPStreamParameters

OBJECT IDENTIFIER ::= { ieee8021PSFPObjects 4 }

-- =====
-- The ieee8021PSFPStreamFilterParameters subtree
-- This subtree defines the objects necessary for the management
-- of the stream filters for IEEE Std 802.1Q.
-- =====

-- =====
-- the ieee8021PSFPStreamFilterTable
-- =====

ieee8021PSFPStreamFilterTable OBJECT-TYPE

SYNTAX SEQUENCE OF Ieee8021PSFPStreamFilterEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"A table that contains the per-filter instance
manageable parameters for stream filters.

A row in the table exists for each stream filter instance.
associated with a Bridge component.

All writable objects in this table must be
persistent over power up restart/reboot."

```
REFERENCE    "8.6.5, 8.6.5.1 12.31.2"
::= { ieee8021PSFPStreamFilterParameters 1 }

ieee8021PSFPStreamFilterEntry OBJECT-TYPE
SYNTAX      Ieee8021PSFPStreamFilterEntry
MAX-ACCESS  not-accessible
STATUS      current
DESCRIPTION
    "A list of objects that contains the manageable parameters for
    stream filters for a Bridge component."
INDEX       { ieee8021BridgeBaseComponentId,
               ieee8021PSFPStreamFilterInstance
             }
::= { ieee8021PSFPStreamFilterTable 1 }

Ieee8021PSFPStreamFilterEntry ::=
SEQUENCE {
    ieee8021PSFPStreamFilterInstance
        Unsigned32,
    ieee8021PSFPStreamHandleSpec
        Integer32,
    ieee8021PSFPPrioritySpec
        Integer32,
    ieee8021PSFPStreamGateInstanceID
        Unsigned32,
    ieee8021PSFPFilterSpecificationList
        OCTET STRING,
    ieee8021PSFPMatchingFramesCount
        Counter64,
    ieee8021PSFPPassingFramesCount
        Counter64,
    ieee8021PSFPNotPassingFramesCount
        Counter64,
    ieee8021PSFPPassingSDUCount
        Counter64,
    ieee8021PSFPNotPassingSDUCount
        Counter64,
    ieee8021PSFPREDFramesCount
        Counter64,
    ieee8021PSFPStreamBlockedDueToOversizeFrameEnable
        TruthValue,
    ieee8021PSFPStreamBlockedDueToOversizeFrame
        TruthValue,
    ieee8021PSFPStreamFilterEntryRowStatus
        RowStatus
}

ieee8021PSFPStreamFilterInstance OBJECT-TYPE
SYNTAX      Unsigned32
MAX-ACCESS  not-accessible
STATUS      current
DESCRIPTION
    "The StreamFilterInstance parameter is an index into the
    StreamFilterTable.

    The value of this object MUST be retained across
    reinitializations of the management system."
REFERENCE    "8.6.5.1, 12.31.2"
::= { ieee8021PSFPStreamFilterEntry 1 }
```

ieee8021PSFPStreamHandleSpec OBJECT-TYPE
SYNTAX Integer32 (-1..2147483647)
MAX-ACCESS read-create
STATUS current
DESCRIPTION
 "The StreamHandleSpec parameter contains a stream identifier specification value. A value of -1 denotes the wild card value; all positive values denote stream identifier values.

 The value of this object MUST be retained across reinitializations of the management system."
REFERENCE "8.6.5.1, 12.31.2"
 ::= { ieee8021PSFPStreamFilterEntry 2}

ieee8021PSFPPrioritySpec OBJECT-TYPE
SYNTAX Integer32 (-1..2147483647)
MAX-ACCESS read-create
STATUS current
DESCRIPTION
 "The PrioritySpec parameter contains a priority specification value. A value of -1 denotes the wild card value; zero or positive values denote priority values.

 The value of this object MUST be retained across reinitializations of the management system."
REFERENCE "8.6.5.1, 12.31.2"
 ::= { ieee8021PSFPStreamFilterEntry 3}

ieee8021PSFPStreamGateInstanceID OBJECT-TYPE
SYNTAX Unsigned32
MAX-ACCESS read-create
STATUS current
DESCRIPTION
 "The StreamGateInstance parameter contains the index of an entry in the Stream Gate Table.

 The value of this object MUST be retained across reinitializations of the management system."
REFERENCE "8.6.5.1, 12.31.2"
 ::= { ieee8021PSFPStreamFilterEntry 4}

ieee8021PSFPFilterSpecificationList OBJECT-TYPE
SYNTAX OCTET STRING
MAX-ACCESS read-create
STATUS current
DESCRIPTION
 "The FilterSpecificationList parameter contains a list of filter specifications associated with this stream filter.

 The octet string value represents the contents of the list as an ordered list of entries, each encoded as a TLV, as follows.

 The first octet of each TLV is interpreted as an unsigned integer representing a filter specification type:
 0: Maximum SDU Size.
 1: Flow meter instance identifier.
 2-255: Reserved for future gate operations

The second and third octets of the TLV are the length field, interpreted as an unsigned integer, indicating the number of octets of the value that follows the length. A length of zero indicates that there is no value (i.e., the filter specification has no parameters).

The fourth through (4 + length - 1)th octets encode the parameters of the filter specification, as defined for each filter specification type.

- Maximum SDU Size:
A single SDU size parameter is encoded in four octets, and is interpreted as an unsigned integer value.
- Flow meter instance identifier:
A single flow meter instance identifier is encoded in four octets, and is interpreted as an unsigned integer value.

The value of this object MUST be retained across reinitializations of the management system."

REFERENCE "8.6.5.1, 12.31.2"
::= { ieee8021PSFPStreamFilterEntry 5}

ieee8021PSFPMatchingFramesCount OBJECT-TYPE
SYNTAX Counter64
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"The MatchingFramesCount counter counts received frames that match this stream filter."
REFERENCE "8.6.5.1, 12.31.2"
::= { ieee8021PSFPStreamFilterEntry 6}

ieee8021PSFPPassingFramesCount OBJECT-TYPE
SYNTAX Counter64
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"The PassingFramesCount counter counts received frames that pass the gate associated with this stream filter."
REFERENCE "8.6.5.1, 12.31.2"
::= { ieee8021PSFPStreamFilterEntry 7}

ieee8021PSFPNotPassingFramesCount OBJECT-TYPE
SYNTAX Counter64
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"The NotPassingFramesCount counter counts received frames that do not pass the gate associated with this stream filter."
REFERENCE "8.6.5.1, 12.31.2"
::= { ieee8021PSFPStreamFilterEntry 8}

ieee8021PSFPPassingSDUCount OBJECT-TYPE
SYNTAX Counter64

```
MAX-ACCESS    read-only
STATUS        current
DESCRIPTION
    "The PassingSDUCount counter counts received frames that
    pass the SDU size filter specification associated
    with this stream filter.
    "
REFERENCE     "8.6.5.1, 12.31.2"
::= { ieee8021PSFPStreamFilterEntry 9}

ieee8021PSFPNotPassingSDUCount OBJECT-TYPE
SYNTAX        Counter64
MAX-ACCESS    read-only
STATUS        current
DESCRIPTION
    "The NotPassingSDUCount counter counts received frames that
    do not pass the SDU size filter specification associated
    with this stream filter.
    "
REFERENCE     "8.6.5.1, 12.31.2"
::= { ieee8021PSFPStreamFilterEntry 10}

ieee8021PSFPPREDFramesCount OBJECT-TYPE
SYNTAX        Counter64
MAX-ACCESS    read-only
STATUS        current
DESCRIPTION
    "The REDFramesCount counter counts received
    random early detection (RED) frames associated
    with this stream filter.
    "
REFERENCE     "8.6.5.1, 12.31.2"
::= { ieee8021PSFPStreamFilterEntry 11}

ieee8021PSFPStreamBlockedDueToOversizeFrameEnable OBJECT-TYPE
SYNTAX        TruthValue
MAX-ACCESS    read-create
STATUS        current
DESCRIPTION
    "The ieee8021PSFPStreamBlockedDueToOversizeFrameEnable object
    contains a Boolean value that indicates whether the
    ieee8021PSFPStreamBlockedDueToOversizeFrame function is
    enabled (TRUE) or disabled (FALSE).

    The value of this object MUST be retained across
    reinitializations of the management system."
REFERENCE     "8.6.5.1, 8.6.5.1.1, 12.31.2"
DEFVAL { false }
::= { ieee8021PSFPStreamFilterEntry 12 }

ieee8021PSFPStreamBlockedDueToOversizeFrame OBJECT-TYPE
SYNTAX        TruthValue
MAX-ACCESS    read-create
STATUS        current
DESCRIPTION
    "The ieee8021PSFPStreamBlockedDueToOversizeFrame object
    contains a Boolean value that indicates whether, if the
    ieee8021PSFPStreamBlockedDueToOversizeFrame function is
    enabled, all frames are to be discarded (TRUE)
```

or not (FALSE).

The value of this object MUST be retained across
reinitializations of the management system."

REFERENCE "8.6.5.1, 8.6.5.1.1, 12.31.2"

DEFVAL { false }

::= { ieee8021PSFPStreamFilterEntry 13 }

ieee8021PSFPStreamFilterEntryRowStatus OBJECT-TYPE

SYNTAX RowStatus

MAX-ACCESS read-create

STATUS current

DESCRIPTION

"The status of the row.

The writable columns in a row can not be changed if the row
is active. All columns MUST have a valid value before a row
can be activated.

"

::= { ieee8021PSFPStreamFilterEntry 14 }

-- =====
-- The ieee8021PSFPStreamGateParameters subtree
-- This subtree defines the objects necessary for the management
-- of the stream gate scheduling mechanism for IEEE Std 802.1Q.
-- =====

-- =====
-- the ieee8021PSFPStreamGateTable
-- =====

ieee8021PSFPStreamGateTable OBJECT-TYPE

SYNTAX SEQUENCE OF Ieee8021PSFPStreamGateEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"A table that contains the per-gate instance
manageable parameters for stream gate scheduling.

For a given Bridge component, a row in the table exists for
each stream gate instance.

All writable objects in this table must be
persistent over power up restart/reboot."

REFERENCE "8.6.8.4, 8.6.9.4, 12.31.3"

::= { ieee8021PSFPStreamGateParameters 1 }

ieee8021PSFPStreamGateEntry OBJECT-TYPE

SYNTAX Ieee8021PSFPStreamGateEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"A list of objects that contains the manageable parameters for
stream gate scheduling for a Bridge component."

INDEX { ieee8021BridgeBaseComponentId,
ieee8021PSFPStreamGateInstance
}

::= { ieee8021PSFPStreamGateTable 1 }

```
Ieee8021PSFPStreamGateEntry ::=
    SEQUENCE {
        ieee8021PSFPStreamGateInstance
            Unsigned32,
        ieee8021PSFPGateEnabled
            TruthValue,
        ieee8021PSFPAdminGateStates
            INTEGER,
        ieee8021PSFPOperGateStates
            INTEGER,
        ieee8021PSFPAdminControlListLength
            Unsigned32,
        ieee8021PSFPOperControlListLength
            Unsigned32,
        ieee8021PSFPAdminControlList
            OCTET STRING,
        ieee8021PSFPOperControlList
            OCTET STRING,
        ieee8021PSFPAdminCycleTimeNumerator
            Unsigned32,
        ieee8021PSFPAdminCycleTimeDenominator
            Unsigned32,
        ieee8021PSFPOperCycleTimeNumerator
            Unsigned32,
        ieee8021PSFPOperCycleTimeDenominator
            Unsigned32,
        ieee8021PSFPAdminCycleTimeExtension
            Unsigned32,
        ieee8021PSFPOperCycleTimeExtension
            Unsigned32,
        ieee8021PSFPAdminBaseTime
            IEEE8021STPTPtimeValue,
        ieee8021PSFPOperBaseTime
            IEEE8021STPTPtimeValue,
        ieee8021PSFPConfigChange
            TruthValue,
        ieee8021PSFPConfigChangeTime
            IEEE8021STPTPtimeValue,
        ieee8021PSFPTickGranularity
            Unsigned32,
        ieee8021PSFPCurrentTime
            IEEE8021STPTPtimeValue,
        ieee8021PSFPConfigPending
            TruthValue,
        ieee8021PSFPConfigChangeError
            Counter64,
        ieee8021PSFPAdminIPV
            Integer32,
        ieee8021PSFPOperIPV
            Integer32,
        ieee8021PSFPGateClosedDueToInvalidRxEnable
            TruthValue,
        ieee8021PSFPGateClosedDueToInvalidRx
            TruthValue,
        ieee8021PSFPGateClosedDueToOctetsExceededEnable
            TruthValue,
        ieee8021PSFPGateClosedDueToOctetsExceeded
            TruthValue,
```

```
ieee8021PSFPStreamGateEntryRowStatus
    RowStatus
}

ieee8021PSFPStreamGateInstance OBJECT-TYPE
    SYNTAX      Unsigned32
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "The StreamGateInstance parameter is an index into the
        StreamGateTable.

        The value of this object MUST be retained across
        reinitializations of the management system."
    REFERENCE   "8.6.5.1, 8.6.5.1.2, 12.31.3"
    ::= { ieee8021PSFPStreamGateEntry 1 }

ieee8021PSFPGateEnabled OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-create
    STATUS      current
    DESCRIPTION
        "The GateEnabled parameter determines whether the stream gate
        is active (true) or inactive (false).

        The value of this object MUST be retained across
        reinitializations of the management system."
    REFERENCE   "8.6.8.4, 8.6.9.4, 12.31.3"
    DEFVAL { false }
    ::= { ieee8021PSFPStreamGateEntry 2 }

ieee8021PSFPAdminGateStates OBJECT-TYPE
    SYNTAX      INTEGER { open(1), closed(2) }
    MAX-ACCESS  read-create
    STATUS      current
    DESCRIPTION
        "The administrative value of the GateStates parameter for the
        stream gate.
        The open value indicates that the gate is open,
        the closed value indicates that the gate is closed.

        The value of this object MUST be retained across
        reinitializations of the management system."
    REFERENCE   "8.6.8.4, 8.6.9.4, 12.31.3"
    ::= { ieee8021PSFPStreamGateEntry 3 }

ieee8021PSFPOperGateStates OBJECT-TYPE
    SYNTAX      INTEGER { open(1), closed(2) }
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The operational value of the GateStates parameter for the
        stream gate.
        The open value indicates that the gate is open,
        the closed value indicates that the gate is closed."
    REFERENCE   "8.6.8.4, 8.6.9.4, 12.31.3"
    ::= { ieee8021PSFPStreamGateEntry 4 }
```

ieee8021PSFPAdminControlListLength OBJECT-TYPE

SYNTAX Unsigned32

MAX-ACCESS read-create

STATUS current

DESCRIPTION

"The administrative value of the ListMax parameter for the gate.
The integer value indicates the number of entries (TLVs) in the
AdminControlList.

The value of this object MUST be retained across
reinitializations of the management system."

REFERENCE "8.6.8.4, 8.6.9.4, 12.31.3"

::= { ieee8021PSFPStreamGateEntry 5 }

ieee8021PSFPOperControlListLength OBJECT-TYPE

SYNTAX Unsigned32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The operational value of the ListMax parameter for the gate.
The integer value indicates the number of entries (TLVs) in the
OperControlList."

REFERENCE "8.6.8.4, 8.6.9.4, 12.31.3"

::= { ieee8021PSFPStreamGateEntry 6 }

ieee8021PSFPAdminControlList OBJECT-TYPE

SYNTAX OCTET STRING

MAX-ACCESS read-create

STATUS current

DESCRIPTION

"The administrative value of the ControlList parameter for the gate.
The octet string value represents the contents of the control list as
an ordered list of entries, each encoded as a TLV, as follows.
The first octet of each TLV is interpreted as an
unsigned integer representing a gate operation name:

0: SetGateAndIPV

1-255: Reserved for future gate operations

The second octet of the TLV is the length field,
interpreted as an unsigned integer, indicating the number of
octets of the value that follows the length. A length of
zero indicates that there is no value
(i.e., the gate operation has no parameters).

The third through (3 + length -1)th octets encode the
parameters of the gate operation, in the order that they
appear in the definition of the operation
in Table 8-7. Three parameter types are defined:

- StreamGateState:

A GateState parameter is encoded in a single octet, and
is interpreted as an integer value.
The value 1 indicates open; the value 2 indicates closed.

- IPV:

An IPV is encoded in four octets as a 32-bit
signed integer. A negative denotes the null value;
zero or positive values denote internal priority values.

- TimeInterval:
A TimeInterval is encoded in 4 octets as a 32-bit unsigned integer, representing a number of nanoseconds. The first octet encodes the most significant 8 bits of the integer, and the fourth octet encodes the least significant 8 bits.
- IntervalOctetMax:
An integer representing the maximum number of MSDU octets that are permitted to pass the gate during the specified TimeInterval. If this parameter is omitted, there is no maximum.

The value of this object MUST be retained across reinitializations of the management system."

REFERENCE "8.6.8.4, 8.6.9.4, 12.31.3"
::= { ieee8021PSFPStreamGateEntry 7 }

ieee8021PSFPOperControlList OBJECT-TYPE

SYNTAX OCTET STRING

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The operational value of the ControlList parameter for the gate. The octet string value represents the contents of the control list as an ordered list of entries, each encoded as a TLV, as follows. The first octet of each TLV is interpreted as an unsigned integer representing a gate operation name:
0: SetGateAndIPV
1-255: Reserved for future gate operations

The second octet of the TLV is the length field, interpreted as an unsigned integer, indicating the number of octets of the value that follows the length. A length of zero indicates that there is no value (i.e., the gate operation has no parameters).

The third through (3 + length -1)th octets encode the parameters of the gate operation, in the order that they appear in the definition of the operation in Table 8-7. Three parameter types are defined:

- StreamGateState:
A GateState parameter is encoded in a single octet, and is interpreted as an integer value.
The value 1 indicates open; the value 2 indicates closed.
- IPV:
An IPV is encoded in four octets as a 32-bit signed integer. A negative value denotes the null value; zero and positive values denote internal priority values.
- TimeInterval:
A TimeInterval is encoded in 4 octets as a 32-bit unsigned integer, representing a number of nanoseconds. The first octet encodes the most significant 8 bits of the integer, and the fourth octet encodes the least significant 8 bits.
- IntervalOctetMax:
An integer representing the maximum number of MSDU octets

that are permitted to pass the gate during the specified
TimeInterval. If this parameter is omitted, there is
no maximum.

"

REFERENCE "8.6.8.4, 8.6.9.4, 12.31.3"
::= { ieee8021PSFPStreamGateEntry 8 }

ieee8021PSFPAdminCycleTimeNumerator OBJECT-TYPE

SYNTAX Unsigned32
MAX-ACCESS read-create
STATUS current
DESCRIPTION

"The administrative value of the numerator of the CycleTime
parameter for the gate.
The numerator and denominator together represent the cycle time as
a rational number of seconds.

The value of this object MUST be retained across
reinitializations of the management system."

REFERENCE "8.6.8.4, 8.6.9.4, 12.31.3"
::= { ieee8021PSFPStreamGateEntry 9 }

ieee8021PSFPAdminCycleTimeDenominator OBJECT-TYPE

SYNTAX Unsigned32
MAX-ACCESS read-create
STATUS current
DESCRIPTION

"The administrative value of the denominator of the
CycleTime parameter for the gate.
The numerator and denominator together represent the cycle time as
a rational number of seconds.

The value of this object MUST be retained across
reinitializations of the management system."

REFERENCE "8.6.8.4, 8.6.9.4, 12.31.3"
::= { ieee8021PSFPStreamGateEntry 10 }

ieee8021PSFPOperCycleTimeNumerator OBJECT-TYPE

SYNTAX Unsigned32
MAX-ACCESS read-only
STATUS current
DESCRIPTION

"The operational value of the numerator of the
CycleTime parameter for the gate.
The numerator and denominator together represent the cycle
time as a rational number of seconds."

REFERENCE "8.6.8.4, 8.6.9.4, 12.31.3"
::= { ieee8021PSFPStreamGateEntry 11 }

ieee8021PSFPOperCycleTimeDenominator OBJECT-TYPE

SYNTAX Unsigned32
MAX-ACCESS read-only
STATUS current
DESCRIPTION

"The operational value of the denominator of the
CycleTime parameter for the gate.
The numerator and denominator together represent the
cycle time as a rational number of seconds."

REFERENCE "8.6.8.4, 8.6.9.4, 12.31.3"


```
::= { ieee8021PSFPStreamGateEntry 12 }

ieee8021PSFPAdminCycleTimeExtension OBJECT-TYPE
    SYNTAX      Unsigned32
    UNITS       "nanoseconds"
    MAX-ACCESS  read-create
    STATUS      current
    DESCRIPTION
        "The administrative value of the CycleTimeExtension
        parameter for the gate.
        The value is an unsigned integer number of nanoseconds.

        The value of this object MUST be retained across
        reinitializations of the management system."
    REFERENCE   "8.6.8.4, 8.6.9.4, 12.31.3"
    ::= { ieee8021PSFPStreamGateEntry 13 }

ieee8021PSFPOperCycleTimeExtension OBJECT-TYPE
    SYNTAX      Unsigned32
    UNITS       "nanoseconds"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The operational value of the CycleTimeExtension
        parameter for the gate.
        The value is an unsigned integer number of nanoseconds."
    REFERENCE   "8.6.8.4, 8.6.9.4, 12.31.3"
    ::= { ieee8021PSFPStreamGateEntry 14 }

ieee8021PSFPAdminBaseTime OBJECT-TYPE
    SYNTAX      IEEE8021STPTPtimeValue
    UNITS       "PTP time"
    MAX-ACCESS  read-create
    STATUS      current
    DESCRIPTION
        "The administrative value of the BaseTime parameter for the gate.
        The value is a representation of a PTPtime value,
        consisting of a 48-bit integer
        number of seconds and a 32-bit integer number of nanoseconds.

        The value of this object MUST be retained across
        reinitializations of the management system."
    REFERENCE   "8.6.8.4, 8.6.9.4, 12.31.3"
    ::= { ieee8021PSFPStreamGateEntry 15 }

ieee8021PSFPOperBaseTime OBJECT-TYPE
    SYNTAX      IEEE8021STPTPtimeValue
    UNITS       "PTP time"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The operational value of the BaseTime parameter for the gate.
        The value is a representation of a PTPtime value,
        consisting of a 48-bit integer
        number of seconds and a 32-bit integer number of nanoseconds."
    REFERENCE   "8.6.8.4, 8.6.9.4, 12.31.3"
    ::= { ieee8021PSFPStreamGateEntry 16 }

ieee8021PSFPConfigChange OBJECT-TYPE
```

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Amendment 28: Per-Stream Filtering and Policing

SYNTAX TruthValue
MAX-ACCESS read-create
STATUS current
DESCRIPTION
 "The ConfigChange parameter signals the start of a
 configuration change for the gate
 when it is set to TRUE. This should only be done
 when the various administrative parameters
 are all set to appropriate values."
REFERENCE "8.6.8.4, 8.6.9.4, 12.31.3"
::= { ieee8021PSFPStreamGateEntry 17 }

ieee8021PSFPConfigChangeTime OBJECT-TYPE

SYNTAX IEEE8021STPTptimeValue
UNITS "PTP time"
MAX-ACCESS read-only
STATUS current
DESCRIPTION
 "The PTPtime at which the next config change is scheduled to occur.
 The value is a representation of a PTPtime value,
 consisting of a 48-bit integer
 number of seconds and a 32-bit integer number of nanoseconds.

 The value of this object MUST be retained across
 reinitializations of the management system."
REFERENCE "8.6.8.4, 8.6.9.4, 12.31.3"
::= { ieee8021PSFPStreamGateEntry 18 }

ieee8021PSFPTickGranularity OBJECT-TYPE

SYNTAX Unsigned32
MAX-ACCESS read-only
STATUS current
DESCRIPTION
 "The granularity of the cycle time clock, represented as an
 unsigned number of tenths of nanoseconds.

 The value of this object MUST be retained across
 reinitializations of the management system."
REFERENCE "8.6.8.4, 8.6.9.4, 12.31.3"
::= { ieee8021PSFPStreamGateEntry 19 }

ieee8021PSFPCurrentTime OBJECT-TYPE

SYNTAX IEEE8021STPTptimeValue
MAX-ACCESS read-only
STATUS current
DESCRIPTION
 "The current time, in PTPtime, as maintained by the local system.
 The value is a representation of a PTPtime value,
 consisting of a 48-bit integer
 number of seconds and a 32-bit integer number of nanoseconds."
REFERENCE "8.6.8.4, 8.6.9.4, 12.31.3"
::= { ieee8021PSFPStreamGateEntry 20 }

ieee8021PSFPConfigPending OBJECT-TYPE

SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current
DESCRIPTION
 "The value of the ConfigPending state machine variable.

The value is TRUE if a configuration change is in progress but has not yet completed."
REFERENCE "8.6.8.4, 8.6.9.4, 12.31.3"
::= { ieee8021PSFPStreamGateEntry 21 }

ieee8021PSFPConfigChangeError OBJECT-TYPE
SYNTAX Counter64
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"A counter of the number of times that a re-configuration of the traffic schedule has been requested with the old schedule still running and the requested base time was in the past."
REFERENCE "8.6.8.4, 8.6.9.3, 8.6.9.1.1, 12.31.3"
::= { ieee8021PSFPStreamGateEntry 23 }

ieee8021PSFPAdminIPV OBJECT-TYPE
SYNTAX Integer32 (-1..2147483647)
MAX-ACCESS read-create
STATUS current
DESCRIPTION
"The administrative value of the IPV parameter for the gate. A value of -1 denotes the null value."
REFERENCE "8.6.5.1.2, 8.6.10, 12.31.3"
::= { ieee8021PSFPStreamGateEntry 24 }

ieee8021PSFPOperIPV OBJECT-TYPE
SYNTAX Integer32 (-1..2147483647)
MAX-ACCESS read-create
STATUS current
DESCRIPTION
"The operational value of the IPV parameter for the gate. A value of -1 denotes the null value."
REFERENCE "8.6.5.1.2, 8.6.10, 12.31.3"
::= { ieee8021PSFPStreamGateEntry 25 }

ieee8021PSFPGateClosedDueToInvalidRxEnable OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-create
STATUS current
DESCRIPTION
"The PSFPGateClosedDueToInvalidRxEnable object contains a Boolean value that indicates whether the PSFPGateClosedDueToInvalidRx function is enabled (TRUE) or disabled (FALSE).

The value of this object MUST be retained across reinitializations of the management system."
REFERENCE "8.6.5.1, 8.6.5.1.2, 12.31.3"
DEFVAL { false }
::= { ieee8021PSFPStreamGateEntry 26 }

ieee8021PSFPGateClosedDueToInvalidRx OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-create
STATUS current

DESCRIPTION

"The PSFPGateClosedDueToInvalidRx object contains a Boolean value that indicates whether, if the PSFPGateClosedDueToInvalidRx function is enabled, all frames are to be discarded (TRUE) or not (FALSE).

The value of this object MUST be retained across reinitializations of the management system."

REFERENCE "8.6.5.1, 8.6.5.1.2, 12.31.3"

DEFVAL { false }

::= { ieee8021PSFPStreamGateEntry 27}

ieee8021PSFPGateClosedDueToOctetsExceededEnable OBJECT-TYPE

SYNTAX TruthValue

MAX-ACCESS read-create

STATUS current

DESCRIPTION

"The PSFPGateClosedDueToOctetsExceededEnable object contains a Boolean value that indicates whether the PSFPGateClosedDueToOctetsExceeded function is enabled (TRUE) or disabled (FALSE).

The value of this object MUST be retained across reinitializations of the management system."

REFERENCE "8.6.5.1, 8.6.5.1.2, 12.31.3"

DEFVAL { false }

::= { ieee8021PSFPStreamGateEntry 28}

ieee8021PSFPGateClosedDueToOctetsExceeded OBJECT-TYPE

SYNTAX TruthValue

MAX-ACCESS read-create

STATUS current

DESCRIPTION

"The PSFPGateClosedDueToOctetsExceeded parameter contains a Boolean value that indicates whether, if the PSFPGateClosedDueToOctetsExceeded function is enabled, all frames are to be discarded (TRUE) or not (FALSE).

The value of this object MUST be retained across reinitializations of the management system."

REFERENCE "8.6.5.1, 8.6.5.1.2, 12.31.3"

DEFVAL { false }

::= { ieee8021PSFPStreamGateEntry 29}

ieee8021PSFPStreamGateEntryRowStatus OBJECT-TYPE

SYNTAX RowStatus

MAX-ACCESS read-create

STATUS current

DESCRIPTION

"The status of the row.

The writable columns in a row can not be changed if the row is active. All columns MUST have a valid value before a row can be activated.

"

::= { ieee8021PSFPStreamGateEntry 30 }

```
-- =====
-- The ieee8021PSFPFlowMeterParameters subtree
-- This subtree defines the objects necessary for the management
-- of the flow meters for IEEE Std 802.1Q.
-- =====

-- =====
-- the ieee8021PSFPFlowMeterTable
-- =====

ieee8021PSFPFlowMeterTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF Ieee8021PSFPFlowMeterEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "A table that contains the per-meter instance
        manageable parameters for flow meters.

        For a given Bridge component, a row in the table exists for
        each flow meter instance.

        All writable objects in this table must be
        persistent over power up restart/reboot."
    REFERENCE   "8.6.5, 8.6.5.1 12.31.4"
    ::= { ieee8021PSFPFlowMeterParameters 1 }

ieee8021PSFPFlowMeterEntry OBJECT-TYPE
    SYNTAX      Ieee8021PSFPFlowMeterEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "A list of objects that contains the manageable parameters for
        flow meters for a Bridge component."
    INDEX       { ieee8021BridgeBaseComponentId,
                  ieee8021PSFPFlowMeterInstance
                }
    ::= { ieee8021PSFPFlowMeterTable 1 }

Ieee8021PSFPFlowMeterEntry ::=
    SEQUENCE {
        ieee8021PSFPFlowMeterInstance
            Unsigned32,
        ieee8021PSFPFlowMeterCIR
            Unsigned32,
        ieee8021PSFPFlowMeterCBS
            Unsigned32,
        ieee8021PSFPFlowMeterEIR
            Unsigned32,
        ieee8021PSFPFlowMeterEBS
            Unsigned32,
        ieee8021PSFPFlowMeterCF
            Integer32,
        ieee8021PSFPFlowMeterCM
            INTEGER,
        ieee8021PSFPFlowMeterDropOnYellow
            TruthValue,
        ieee8021PSFPFlowMeterMarkAllFramesRedEnable
            TruthValue,
```

```
ieee8021PSFPFlowMeterMarkAllFramesRed
    TruthValue,
ieee8021PSFPFlowMeterEntryRowStatus
    RowStatus
}

ieee8021PSFPFlowMeterInstance OBJECT-TYPE
    SYNTAX      Unsigned32
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "The FlowMeterInstance parameter is an index into the
        FlowMeterTable.

        The value of this object MUST be retained across
        reinitializations of the management system."
    REFERENCE   "8.6.5.1, 12.31.4"
    ::= { ieee8021PSFPFlowMeterEntry 1}

ieee8021PSFPFlowMeterCIR OBJECT-TYPE
    SYNTAX      Unsigned32
    MAX-ACCESS  read-create
    STATUS      current
    DESCRIPTION
        "The FlowMeterCIR parameter contains an integer value that
        represents the CIR value for the flow meter, in bit/s.

        The value of this object MUST be retained across
        reinitializations of the management system."
    REFERENCE   "8.6.5.1, 12.31.4"
    ::= { ieee8021PSFPFlowMeterEntry 2}

ieee8021PSFPFlowMeterCBS OBJECT-TYPE
    SYNTAX      Unsigned32
    MAX-ACCESS  read-create
    STATUS      current
    DESCRIPTION
        "The FlowMeterCBS parameter contains an integer value that
        represents the CBS value for the flow meter, in octets.

        The value of this object MUST be retained across
        reinitializations of the management system."
    REFERENCE   "8.6.5.1, 12.31.4"
    ::= { ieee8021PSFPFlowMeterEntry 3}

ieee8021PSFPFlowMeterEIR OBJECT-TYPE
    SYNTAX      Unsigned32
    MAX-ACCESS  read-create
    STATUS      current
    DESCRIPTION
        "The FlowMeterEIR parameter contains an integer value that
        represents the EIR value for the flow meter, in bit/s.

        The value of this object MUST be retained across
        reinitializations of the management system."
    REFERENCE   "8.6.5.1, 12.31.4"
    ::= { ieee8021PSFPFlowMeterEntry 4}

ieee8021PSFPFlowMeterEBS OBJECT-TYPE
```

SYNTAX Unsigned32
MAX-ACCESS read-create
STATUS current
DESCRIPTION
 "The FlowMeterEBS parameter contains an integer value that represents the EBS value for the flow meter, in octets.

 The value of this object MUST be retained across reinitializations of the management system."
REFERENCE "8.6.5.1, 12.31.4"
::= { ieee8021PSFPFlowMeterEntry 5 }

ieee8021PSFPFlowMeterCF OBJECT-TYPE

SYNTAX Integer32 (0..1)
MAX-ACCESS read-create
STATUS current
DESCRIPTION
 "The FlowMeterCF parameter contains an integer value that represents the CF value for the flow meter, as an integer value 0 or 1.

 The value of this object MUST be retained across reinitializations of the management system."
REFERENCE "8.6.5.1, 12.31.4"
::= { ieee8021PSFPFlowMeterEntry 6 }

ieee8021PSFPFlowMeterCM OBJECT-TYPE

SYNTAX INTEGER {colorBlind(1), colorAware(2)}
MAX-ACCESS read-create
STATUS current
DESCRIPTION
 "The FlowMeterCM parameter contains an integer value that represents the CM value for the flow meter, as an enumerated value indicating colorBlind(1) or colorAware(2).

 The value of this object MUST be retained across reinitializations of the management system."
REFERENCE "8.6.5.1, 12.31.4"
::= { ieee8021PSFPFlowMeterEntry 7 }

ieee8021PSFPFlowMeterDropOnYellow OBJECT-TYPE

SYNTAX TruthValue
MAX-ACCESS read-create
STATUS current
DESCRIPTION
 "The FlowMeterDropOnYellow parameter contains a Boolean value that indicates whether yellow frames are dropped (TRUE) or have drop_eligible set to TRUE (FALSE).

 The value of this object MUST be retained across reinitializations of the management system."
REFERENCE "8.6.5.1, 12.31.4"
::= { ieee8021PSFPFlowMeterEntry 8 }

ieee8021PSFPFlowMeterMarkAllFramesRedEnable OBJECT-TYPE

SYNTAX TruthValue
MAX-ACCESS read-create
STATUS current
DESCRIPTION

"The FlowMeterMarkAllFramesRedEnable parameter contains a Boolean value that indicates whether the MarkAllFramesRed function is enabled (TRUE) or disabled (FALSE).

The value of this object MUST be retained across reinitializations of the management system."

REFERENCE "8.6.5.1, 8.6.5.1.3, 12.31.4"

DEFVAL { false }

::= { ieee8021PSFPFlowMeterEntry 9 }

ieee8021PSFPFlowMeterMarkAllFramesRed OBJECT-TYPE

SYNTAX TruthValue

MAX-ACCESS read-create

STATUS current

DESCRIPTION

"The FlowMeterMarkAllFramesRed parameter contains a Boolean value that indicates whether, if the MarkAllFramesRed function is enabled, all frames are to be discarded (TRUE) or not (FALSE).

The value of this object MUST be retained across reinitializations of the management system."

REFERENCE "8.6.5.1, 8.6.5.1.3, 12.31.4"

DEFVAL { false }

::= { ieee8021PSFPFlowMeterEntry 10 }

ieee8021PSFPFlowMeterEntryRowStatus OBJECT-TYPE

SYNTAX RowStatus

MAX-ACCESS read-create

STATUS current

DESCRIPTION

"The status of the row.

The writable columns in a row can not be changed if the row is active. All columns MUST have a valid value before a row can be activated.

"

::= { ieee8021PSFPFlowMeterEntry 11 }

```
-- =====
-- The ieee8021PSFPStreamParameters subtree
-- This subtree defines the objects necessary for the management
-- of the flow meters for IEEE Std 802.1Q.
-- =====
```

```
-- =====
-- the ieee8021PSFPStreamParameterTable
-- =====
```

ieee8021PSFPStreamParameterTable OBJECT-TYPE

SYNTAX SEQUENCE OF Ieee8021PSFPStreamParameterEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"A table that contains per-Bridge component manageable parameters for PSFP.

A row in the table exists for each Bridge component.

All writable objects in this table must be persistent over power up restart/reboot."

REFERENCE "8.6.5, 8.6.5.1 12.31.4"
::= { ieee8021PSFPStreamParameters 1 }

ieee8021PSFPStreamParameterEntry OBJECT-TYPE
SYNTAX Ieee8021PSFPStreamParameterEntry
MAX-ACCESS not-accessible
STATUS current
DESCRIPTION
"A list of objects that contains the manageable parameters for
flow meters for a Bridge component."
INDEX { ieee8021BridgeBaseComponentId
}
::= { ieee8021PSFPStreamParameterTable 1 }

Ieee8021PSFPStreamParameterEntry ::=
SEQUENCE {
ieee8021PSFPMaxStreamFilterInstances
Unsigned32,
ieee8021PSFPMaxStreamGateInstances
Unsigned32,
ieee8021PSFPMaxFlowMeterInstances
Unsigned32,
ieee8021PSFPSupportedListMax
Unsigned32
}

ieee8021PSFPMaxStreamFilterInstances OBJECT-TYPE
SYNTAX Unsigned32
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"The MaxStreamFilterInstances parameter defines the
maximum number of stream filter instances that are
supported by this Bridge component."
REFERENCE "8.6.5.1, 12.31.2"
::= { ieee8021PSFPStreamParameterEntry 1}

ieee8021PSFPMaxStreamGateInstances OBJECT-TYPE
SYNTAX Unsigned32
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"The MaxStreamGateInstances parameter defines the
maximum number of stream gate instances that are
supported by this Bridge component."
REFERENCE "8.6.5.1, 12.31.3"
::= { ieee8021PSFPStreamParameterEntry 2}

ieee8021PSFPMaxFlowMeterInstances OBJECT-TYPE
SYNTAX Unsigned32
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"The MaxFlowMeterInstances parameter defines the

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IEEE Standard for Local and Metropolitan Area Networks—Bridges and Bridged Networks—
Amendment 28: Per-Stream Filtering and Policing

```
        maximum number of flow meter instances that are
        supported by this Bridge component."
REFERENCE    "8.6.5.1, 12.31.4"
::= { ieee8021PSFPStreamParameterEntry 3}

ieee8021PSFPSupportedListMax OBJECT-TYPE
    SYNTAX      Unsigned32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The SupportedListMax parameter defines the
        The maximum value supported by this Bridge component of
        the AdminControllistLength and
        OperControllistLength parameters."
    REFERENCE   "8.6.5.1, 12.31.3"
    ::= { ieee8021PSFPStreamParameterEntry 4}

-- =====
-- IEEE8021 PSFP MIB - Conformance Information
-- =====

ieee8021PSFPCompliances
    OBJECT IDENTIFIER ::= { ieee8021PSFPConformance 1 }
ieee8021PSFPGroups
    OBJECT IDENTIFIER ::= { ieee8021PSFPConformance 2 }

-- =====
-- units of conformance
-- =====

-- =====
-- the ieee8021PSFPObjectsGroup group
-- =====

ieee8021PSFPObjectsGroup OBJECT-GROUP
    OBJECTS {
        ieee8021PSFPStreamHandleSpec,
        ieee8021PSFPPrioritySpec,
        ieee8021PSFPStreamGateInstanceID,
        ieee8021PSFPFilterSpecificationList,
        ieee8021PSFPMatchingFramesCount,
        ieee8021PSFPPassingFramesCount,
        ieee8021PSFPNotPassingFramesCount,
        ieee8021PSFPPassingSDUCount,
        ieee8021PSFPNotPassingSDUCount,
        ieee8021PSFPREDFramesCount,
        ieee8021PSFPStreamBlockedDueToOversizeFrameEnable,
        ieee8021PSFPStreamBlockedDueToOversizeFrame,
        ieee8021PSFPStreamFilterEntryRowStatus,
        ieee8021PSFPGateEnabled,
        ieee8021PSFPAdminGateStates,
        ieee8021PSFPOperGateStates,
        ieee8021PSFPAdminControllistLength,
        ieee8021PSFPOperControllistLength,
        ieee8021PSFPAdminControllist,
        ieee8021PSFPOperControllist,
        ieee8021PSFPAdminCycleTimeNumerator,
        ieee8021PSFPAdminCycleTimeDenominator,
```

```

ieee8021PSFPOperCycleTimeNumerator,
ieee8021PSFPOperCycleTimeDenominator,
ieee8021PSFPAdminCycleTimeExtension,
ieee8021PSFPOperCycleTimeExtension,
ieee8021PSFPAdminBaseTime,
ieee8021PSFPOperBaseTime,
ieee8021PSFPConfigChange,
ieee8021PSFPConfigChangeTime,
ieee8021PSFPTickGranularity,
ieee8021PSFPCurrentTime,
ieee8021PSFPConfigPending,
ieee8021PSFPConfigChangeError,
ieee8021PSFPAdminIPV,
ieee8021PSFPOperIPV,
ieee8021PSFPGateClosedDueToInvalidRxEnable,
ieee8021PSFPGateClosedDueToInvalidRx,
ieee8021PSFPGateClosedDueToOctetsExceededEnable,
ieee8021PSFPGateClosedDueToOctetsExceeded,
ieee8021PSFPStreamGateEntryRowStatus,
ieee8021PSFPFlowMeterCIR,
ieee8021PSFPFlowMeterCBS,
ieee8021PSFPFlowMeterEIR,
ieee8021PSFPFlowMeterEBS,
ieee8021PSFPFlowMeterCF,
ieee8021PSFPFlowMeterCM,
ieee8021PSFPFlowMeterDropOnYellow,
ieee8021PSFPFlowMeterMarkAllFramesRedEnable,
ieee8021PSFPFlowMeterMarkAllFramesRed,
ieee8021PSFPFlowMeterEntryRowStatus,
ieee8021PSFPMaxStreamFilterInstances,
ieee8021PSFPMaxStreamGateInstances,
ieee8021PSFPMaxFlowMeterInstances,
ieee8021PSFPSupportedListMax
}

```

```

STATUS      current
DESCRIPTION
    "Objects that allow management of PSFP."
 ::= { ieee8021PSFPGroups 1 }

```

```

-- =====
-- compliance statements
-- =====

```

```

ieee8021PSFPCompliance MODULE-COMPLIANCE
    STATUS      current
    DESCRIPTION
        "The compliance statement for devices supporting
        PSFP.

        Support of the objects defined in this MIB module
        also requires support of the IEEE8021-BRIDGE-MIB; the
        provisions of 17.3.2 apply to implementations claiming
        support of this MIB. "

```

```

MODULE -- this module
    MANDATORY-GROUPS {
        ieee8021PSFPObjectsGroup
    }

```

```
}  
  
::= { ieee8021PSFPCompliances 1 }  
  
END
```

Annex A

(normative)

PICS proforma—Bridge implementations⁴

A.5 Major capabilities

Insert the following row at the end of Table A.5:

PSFP	Does the implementation support PSFP?	O	8.6.5.1, 8.6.6.1, 8.6.10, 12.31	Yes []	No []
------	---------------------------------------	---	------------------------------------	---------	--------

A.14 Bridge management

Insert the following row at the end of Table A.14, renumbering item number MGT-249 if necessary:

Item	Feature	Status	References	Support
MGT-249	Does the implementation support the management entities defined in 12.31?	PSFP: M	8.6.5.1, 8.6.6.1, 8.6.10, 12.31	Yes [] N/A []

A.24 Management Information Base (MIB)

Insert the following row at the end of Table A.24, renumbering item number MIB-43 if necessary:

Item	Feature	Status	References	Support
MIB-43	Is the IEEE8021-PSFP-MIB module fully supported (per its MODULE-COMPLIANCE)?	PSFP: O	8.6.5.1, 8.6.6.1, 8.6.10, 12.31, 17.7.24	Yes [] N/A []

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Insert new Table A.46 at the end of Annex A as shown, renumbering as necessary:

A.46 Per-stream filtering and policing

Item	Feature	Status	References	Support
	If per-stream filtering and policing (PSFP in Table A.5) is not supported, mark N/A and ignore the remainder of this table.		8.6.5.1, 8.6.6.1, 8.6.10, 12.31, 17.7.24	N/A []
PSFP1	Does the implementation support the state machines and associated definitions as specified in 8.6.10?	PSFP:M	8.6.5, 8.6.10	Yes [] N/A []
PSFP2	Does the implementation support the management entities defined in 12.31?	PSFP:M	8.6.5.1, 8.6.6.1, 8.6.10, 12.31	Yes [] N/A []
PSFP3	Is the IEEE8021-PSFP-MIB module fully supported (per its MODULE-COMPLIANCE)?	MIB AND PSFP:O	12.31, 17.7.24	Yes [] N/A [] No []

Annex B

(normative)

PICS proforma—End station implementations⁵

B.5 Major capabilities

Insert the following row at the end of Table B.5:

PSFP	Does the implementation support PSFP?	O	8.6.5.1, 8.6.6.1, 8.6.10, 12.31	Yes []	No []
------	---------------------------------------	---	------------------------------------	---------	--------

Insert new Table B.17 at the end of Annex B as shown, renumbering as necessary:

B.17 Per-stream filtering and policing

Item	Feature	Status	References	Support
	If per-stream filtering and policing (PSFP in Table B.5) is not supported, mark N/A and ignore the remainder of this table.		8.6.5.1, 8.6.6.1, 8.6.10, 12.31, 17.7.24	N/A []
PSFP1	Does the implementation support the state machines and associated definitions as specified in 8.6.10?	PSFP:M	8.6.5, 8.6.10	Yes [] N/A []
PSFP2	Does the implementation support the management entities defined in 12.31?	PSFP:M	8.6.5.1, 8.6.6.1, 8.6.10, 12.31	Yes [] N/A []
PSFP3	Is the IEEE8021-PSFP-MIB module fully supported (per its MODULE-COMPLIANCE)?	MIB AND PSFP:O	12.31, 17.7.24	Yes [] N/A [] No []

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Annex U

(informative)

Bibliography

Insert the following entry in the bibliography, renumbering as necessary.

[B1] IEEE Std 802.1Qch, IEEE Standard for Local and metropolitan area networks—Bridges and Bridged Networks—Amendment 29: Cyclic Queuing and Forwarding.

Consensus

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